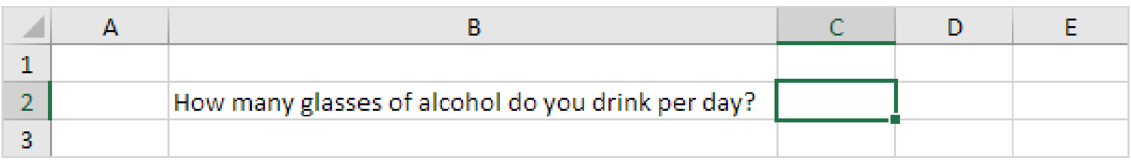
Data Validation

Open MS Excel and do the data validation examples given below. Save your work in one workbook but in a separate worksheet. Rename each worksheet based on the data validation you applied. Name your workbook as Data Validation – [Lastname]. Upload the file in Neo LMS.

Use data validation in Excel to make sure that users enter certain values into a cell.

Data Validation Example

In this example, we restrict users to enter a whole number between 0 and 10.

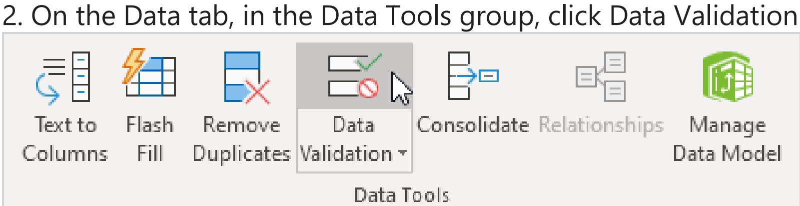


Create Data Validation Rule

To create the data validation rule, execute the following steps.

1. Select cell C2.

2. On the Data tab, in the Data Tools group, click Data Validation.

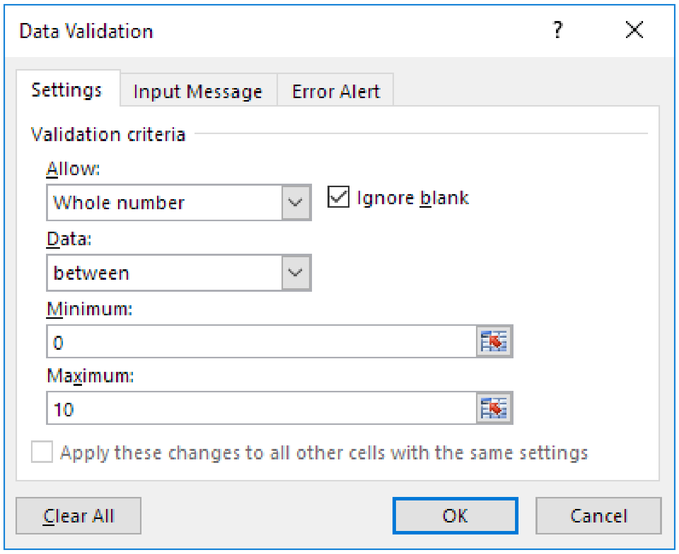


On the Settings tab:

3. In the Allow list, click Whole number.

4. In the Data list, click between.

5. Enter the Minimum and Maximum values.



Input Message

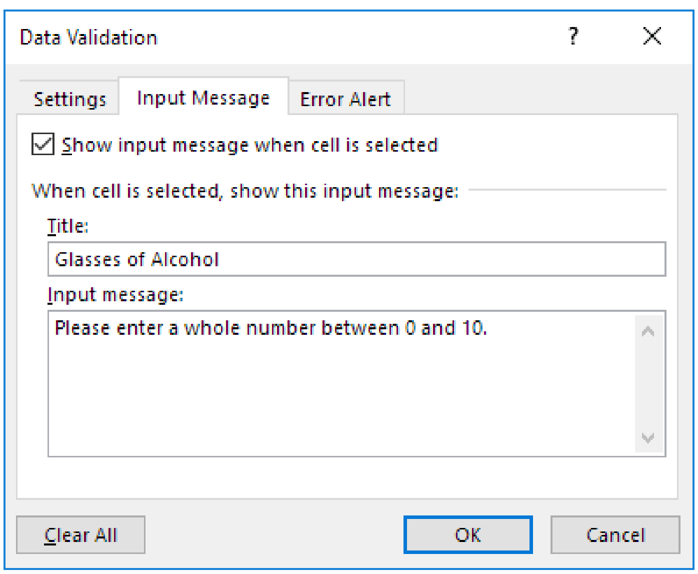
Input messages appear when the user selects the cell and tell the user what to enter.

On the Input Message tab:

1. Check 'Show input message when cell is selected'.

2. Enter a title.

3. Enter an input message.



Error Alert

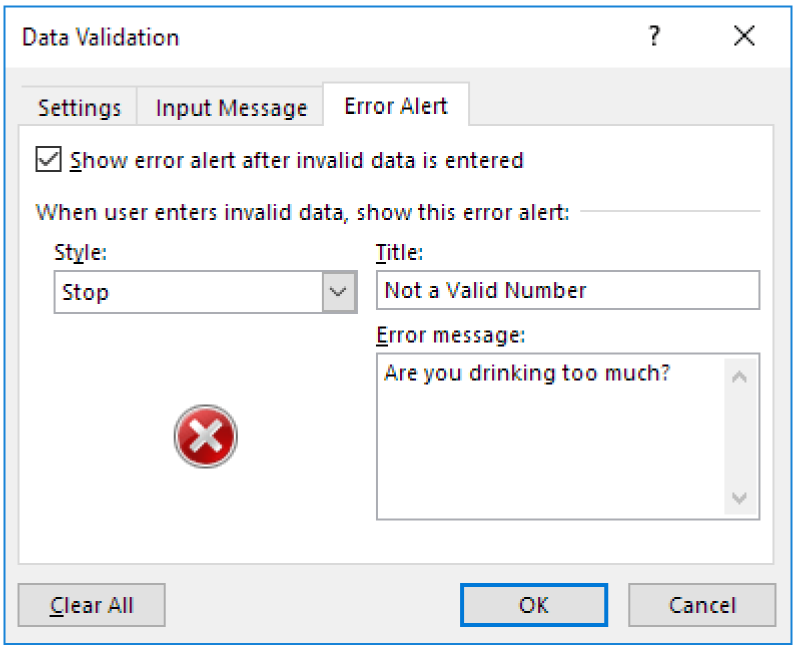
If users ignore the input message and enter a number that is not valid, you can show them an error alert.

On the Error Alert tab:

1. Check 'Show error alert after invalid data is entered'.

2. Enter a title.

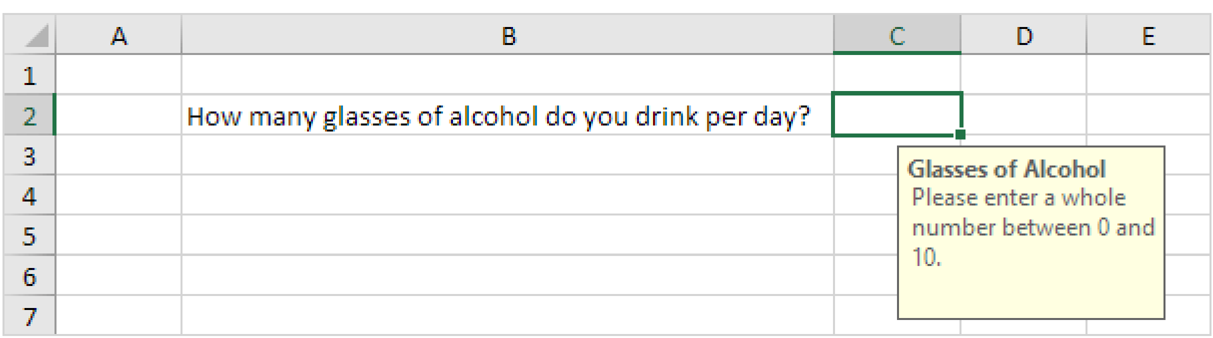
3. Enter an error message.



4. Click OK.

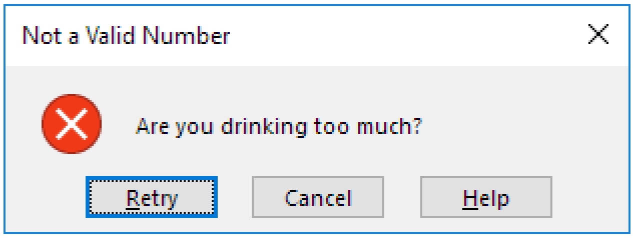
Data Validation Result

1. Select cell C2.



2. Try to enter a number higher than 10.

Result:

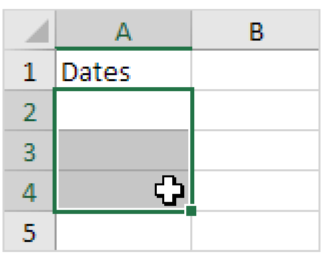


Note: to remove data validation from a cell, select the cell, on the Data tab, in the Data Tools group, click Data Validation, and then click Clear All. You can use Excel's [Go To Special] feature to quickly select all cells with data validation.

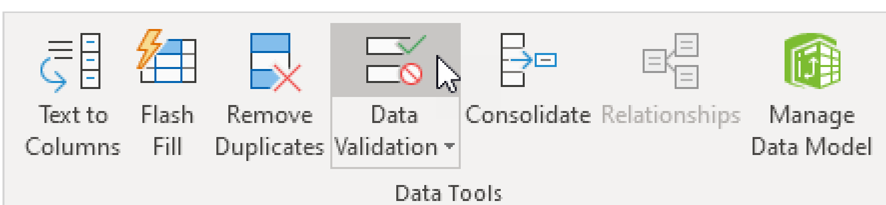
Reject Invalid Dates

This example teaches you how to use data validation to reject invalid dates.

1. Select the range A2:A4.



2. On the Data tab, in the Data Tools group, click Data Validation.

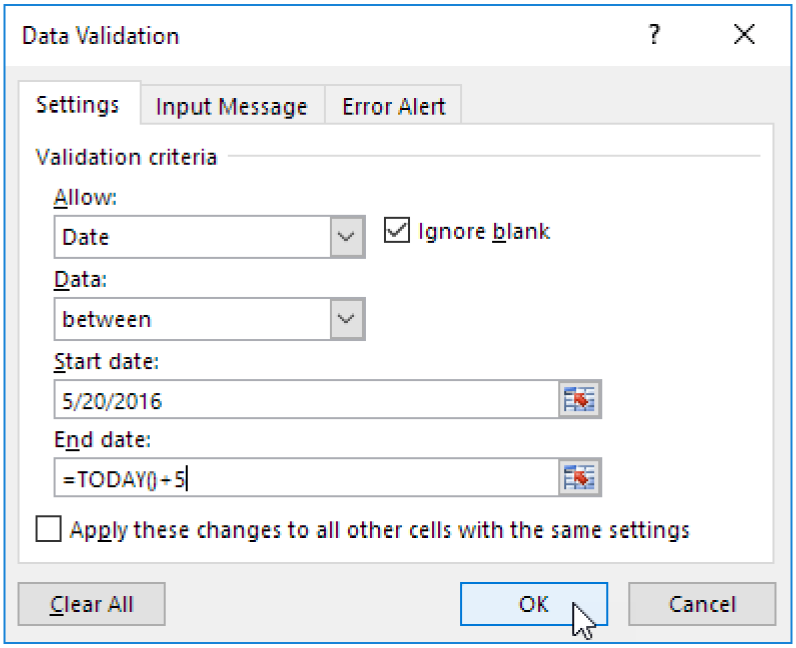


Outside a Date Range

3. In the Allow list, click Date.

4. In the Data list, click between.

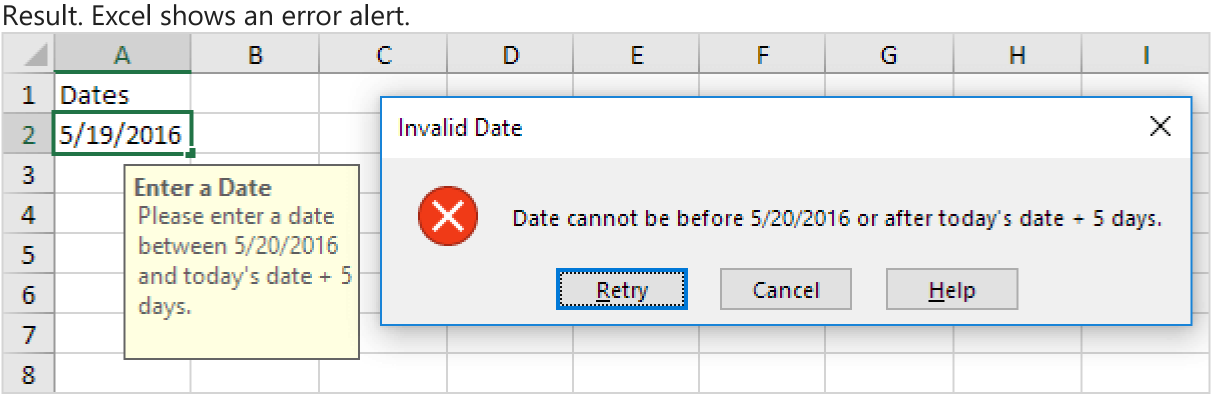
5. Enter the Start date and End date shown below and click OK.



Explanation: all dates between 5/20/2016 and today's date + 5 days are allowed. All dates outside this date range are rejected.

6. Enter the date 5/19/2016 into cell A2.

Result. Excel shows an error alert.

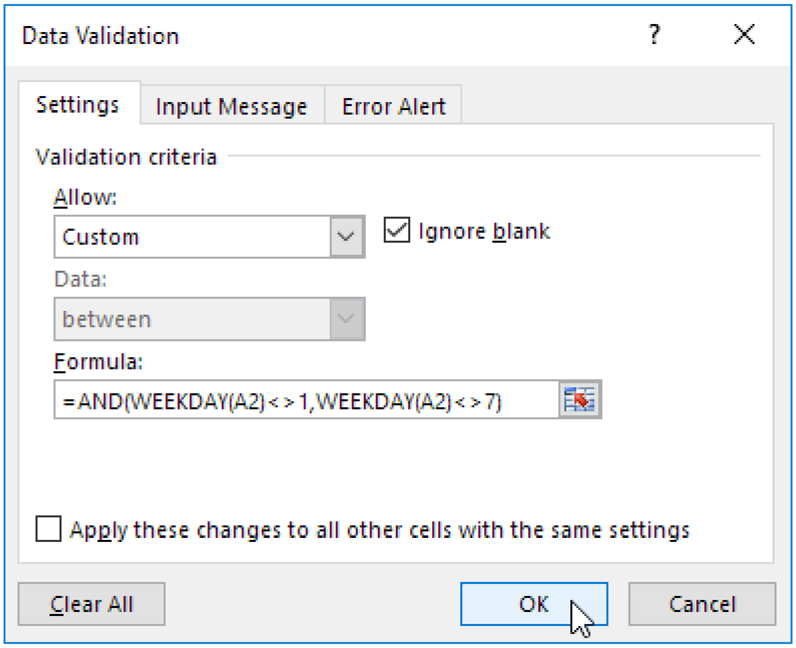


Note: to enter an input message and error alert message, go to the Input Message and Error Alert tab.

Sundays and Saturdays

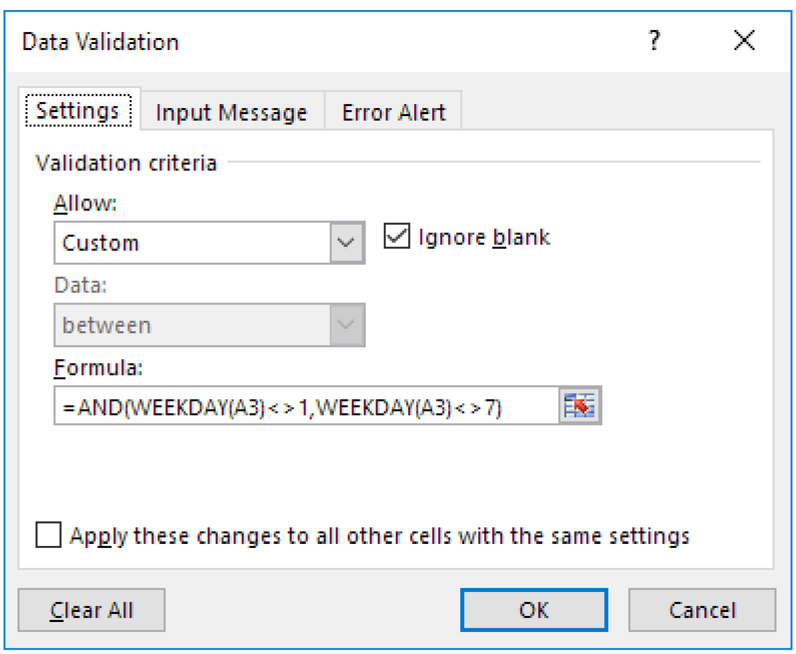
3. In the Allow list, click Custom.

4. In the Formula box, enter the formula shown below and click OK.



Explanation: the WEEKDAY function returns a number from 1 (Sunday) to 7 (Saturday) representing the day of the week of a date. If a date's weekday is not equal to 1 (Sunday) AND not equal to 7 (Saturday), the date is allowed (<> means not equal to). In other words, Mondays, Tuesdays, Wednesdays, Thursdays and Fridays are allowed. Sundays and Saturdays are rejected. Because we selected the range A2:A4 before we clicked on Data Validation, Excel automatically copies the formula to the other cells.

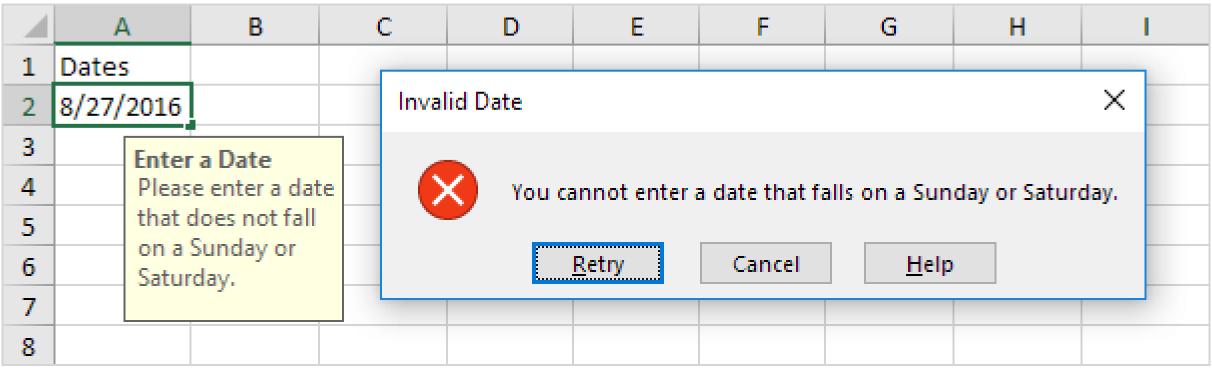
5. To check this, select cell A3 and click Data Validation.



As you can see, this cell also contains the correct formula.

6. Enter the date 8/27/2016 (Saturday) into cell A2.

Result. Excel shows an error alert.

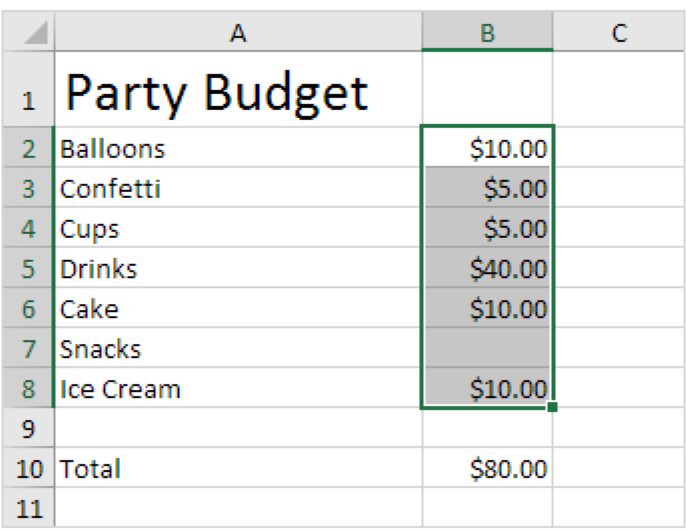


Note: to enter an input message and error alert message, go to the Input Message and Error Alert tab.

Budget Limit

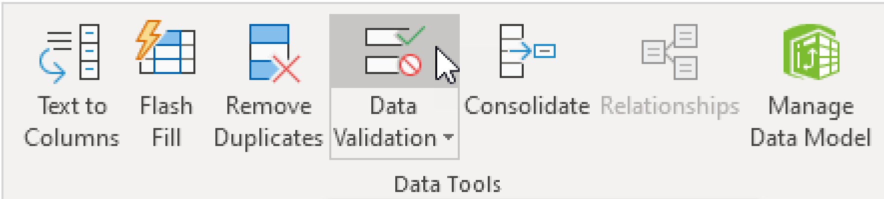
This example teaches you how to use data validation to prevent users from exceeding a budget limit.

1. Select the range B2:B8.



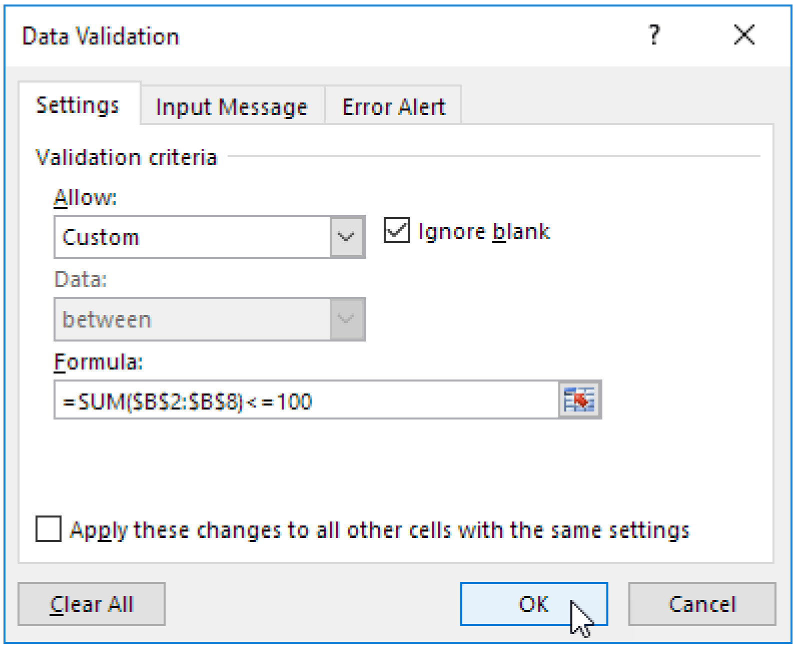
Note: cell B10 contains a SUM function that calculates the sum of the range B2:B8.

2. On the Data tab, in the Data Tools group, click Data Validation.



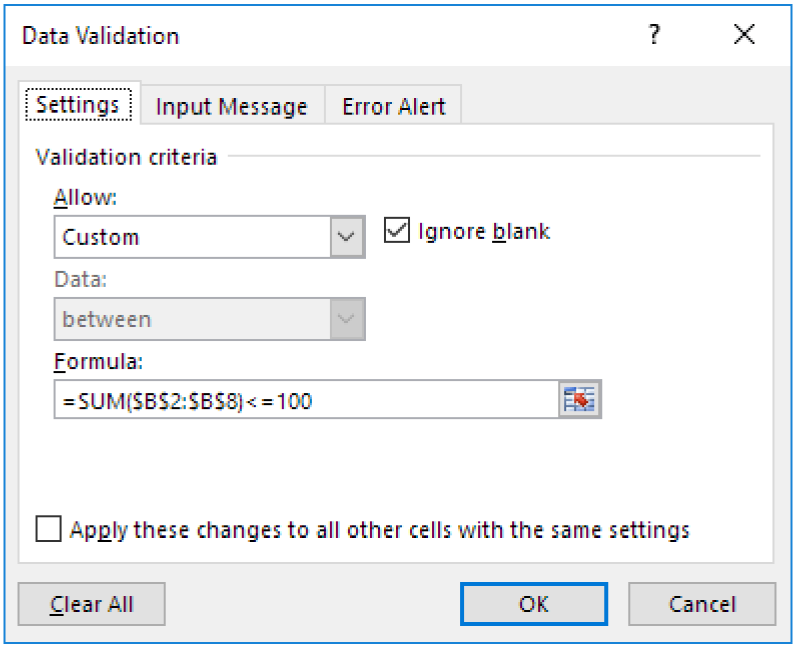
3. In the Allow list, click Custom.

4. In the Formula box, enter the formula shown below and click OK.



Explanation: the sum of the range B2:B8 may not exceed the budget limit of $100. Therefore, we apply data validation to the range B2:B8 (not cell B10!) because this is where the values are entered. Because we selected the range B2:B8 before we clicked on Data Validation, Excel automatically copies the formula to the other cells. Notice how we created an absolute reference ($B$2:$B$8) to fix this reference.

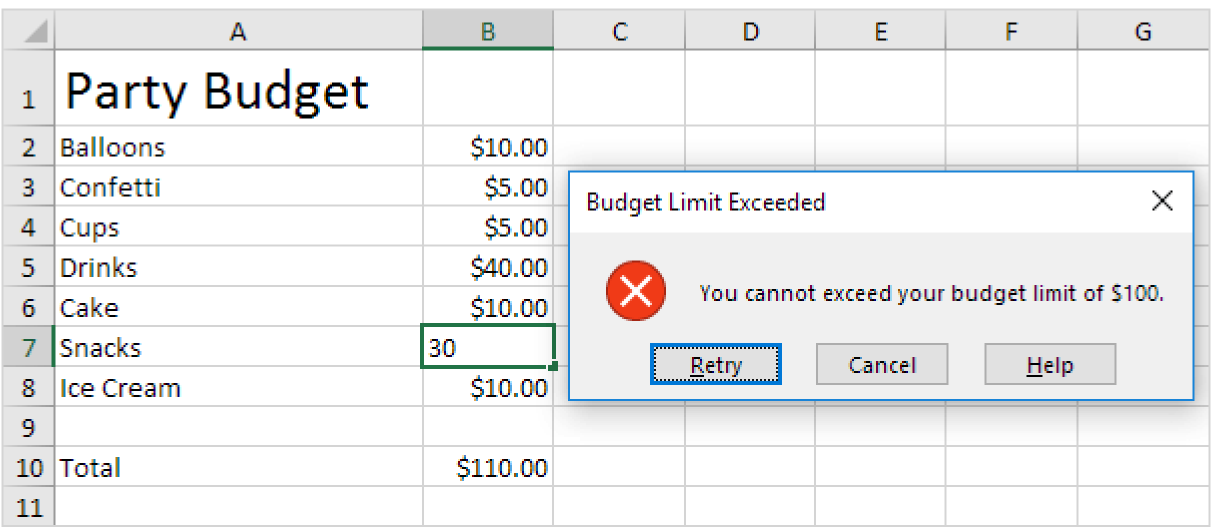
5. To check this, select cell B3 and click Data Validation.



As you can see, this cell also contains the correct formula.

6. Enter the value 30 into cell B7.

Result. Excel shows an error alert. You cannot exceed your budget limit of $100.

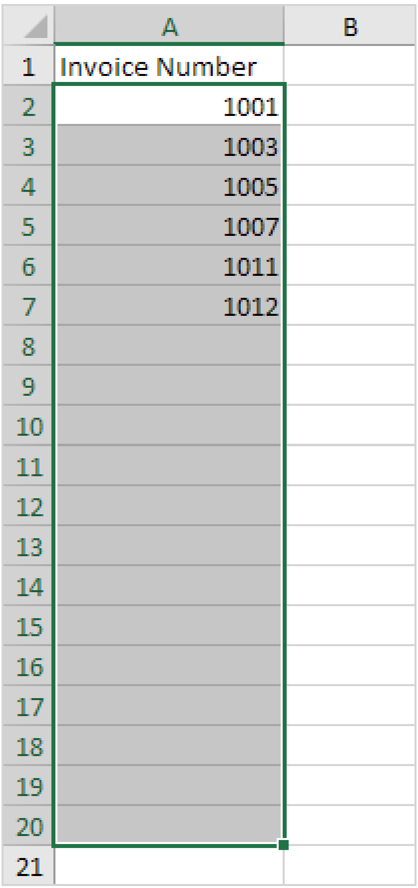


Note: to enter an error alert message, go to the Error Alert tab.

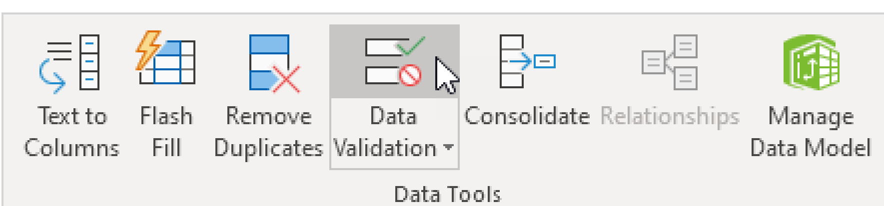
Prevent Duplicate Entries

This example teaches you how to use data validation to prevent users from entering duplicate values.

1. Select the range A2:A20.

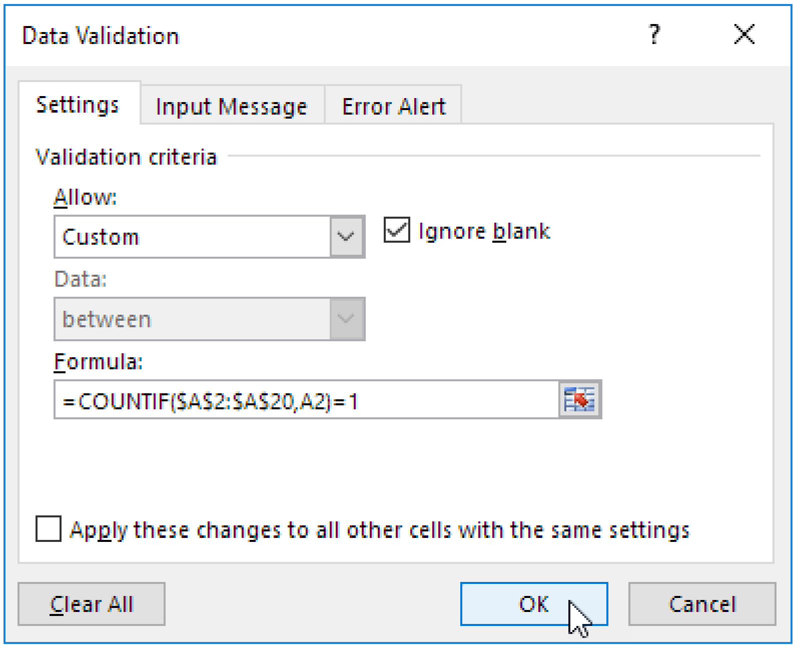


2. On the Data tab, in the Data Tools group, click Data Validation.



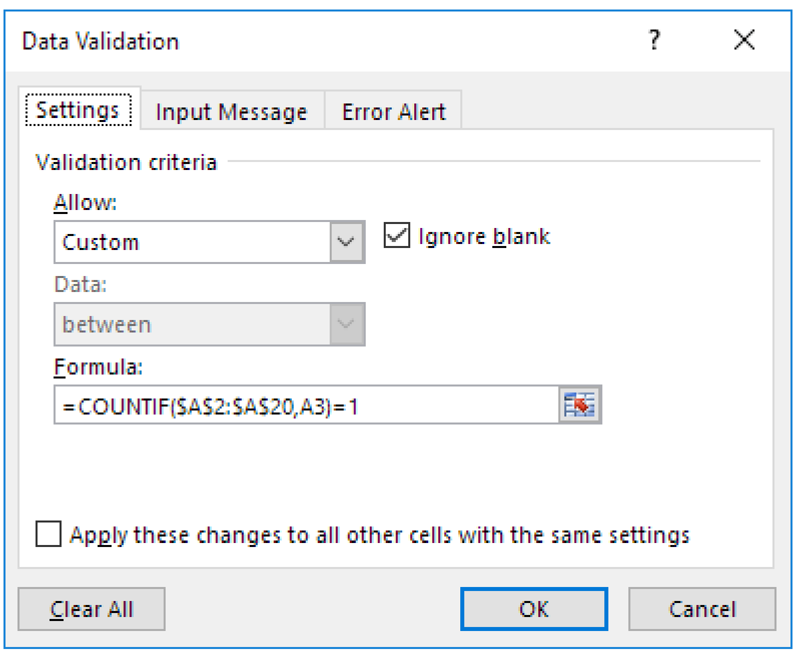
3. In the Allow list, click Custom.

4. In the Formula box, enter the formula shown below and click OK.



Explanation: The [COUNTIF function] takes two arguments. =COUNTIF($A$2:$A$20,A2) counts the number of values in the range A2:A20 that are equal to the value in cell A2. This value may only occur once (=1) since we don't want duplicate entries. Because we selected the range A2:A20 before we clicked on Data Validation, Excel automatically copies the formula to the other cells. Notice how we created an absolute reference ($A$2:$A$20) to fix this reference.

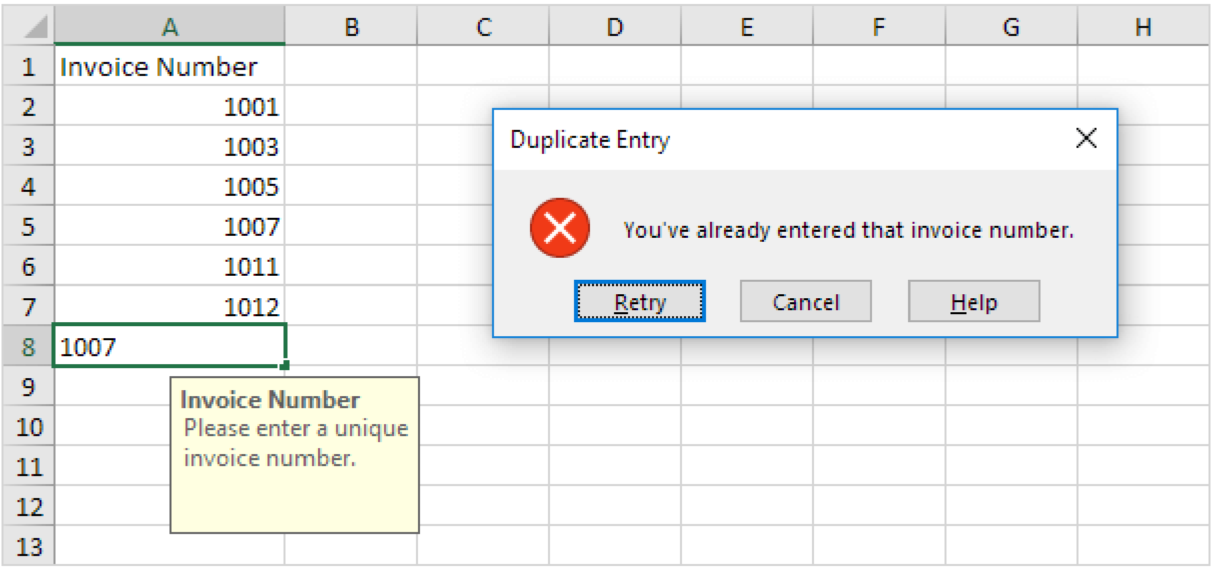
5. To check this, select cell A3 and click Data Validation.



As you can see, this function counts the number of values in the range A2:A20 that are equal to the value in cell A3. Again, this value may only occur once (=1) since we don't want duplicate entries.

6. Enter a duplicate invoice number.

Result. Excel shows an error alert. You've already entered that invoice number.

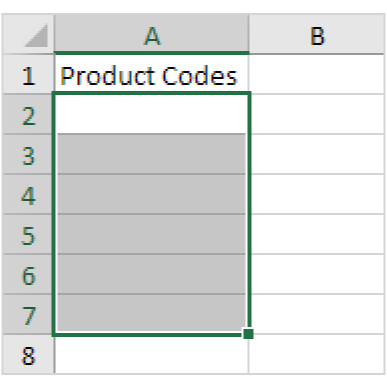


Note: to enter an input message and error alert message, go to the Input Message and Error Alert tab.

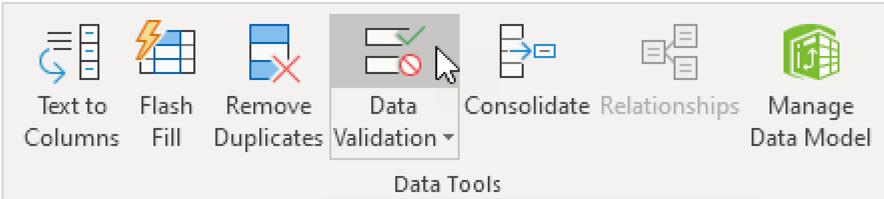
Product Codes

This example teaches you how to use data validation to prevent users from entering incorrect product codes.

1. Select the range A2:A7.

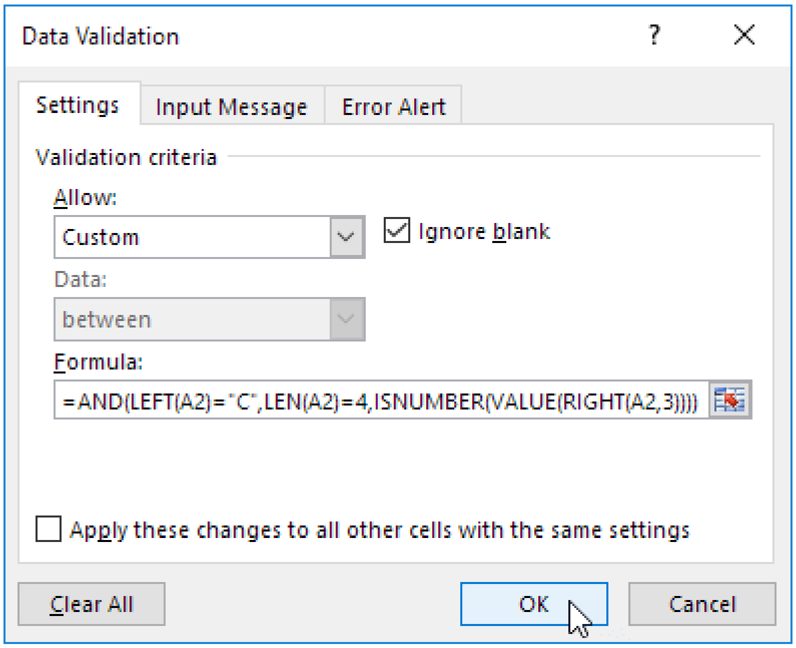


2. On the Data tab, in the Data Tools group, click Data Validation.



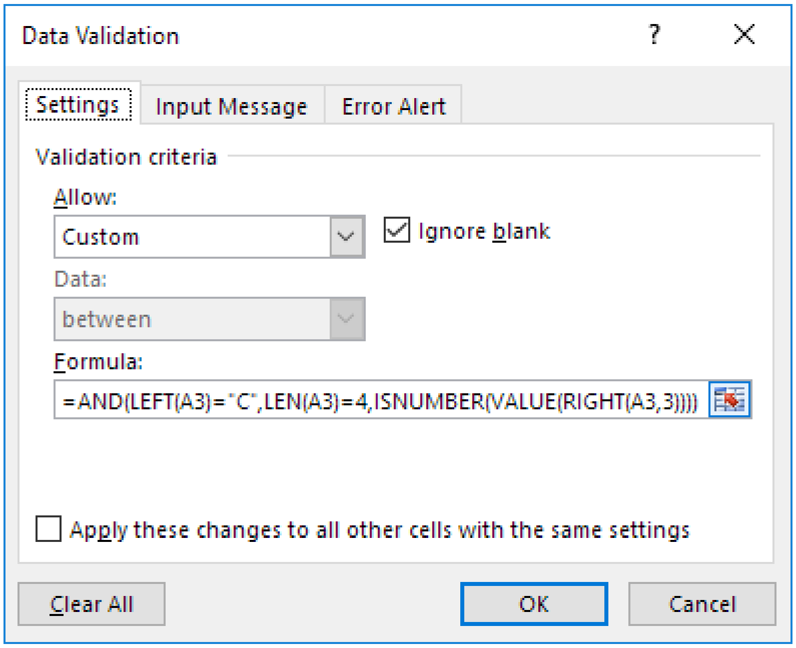
3. In the Allow list, click Custom.

4. In the Formula box, enter the formula shown below and click OK.



Explanation: this AND function has three arguments. LEFT(A2)="C" forces the user to start with the letter C. LEN(A2)=4 forces the user to enter a string with a length of 4 characters. ISNUMBER(VALUE(RIGHT(A2,3))) forces the user to end with 3 numbers. RIGHT(A2,3) extracts the 3 rightmost characters from the text string. The VALUE function converts this text string to a number. ISNUMBER checks whether this value is a number. The AND Function returns TRUE if all conditions are true. Because we selected the range A2:A7 before we clicked on Data Validation, Excel automatically copies the formula to the other cells.

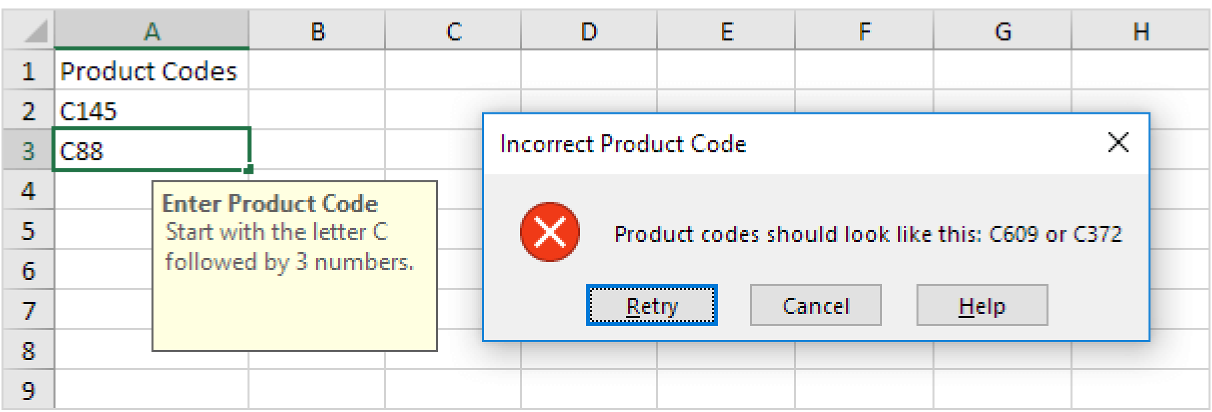
5. To check this, select cell A3 and click Data Validation.



As you can see, this cell also contains the correct formula.

6. Enter an incorrect product code.

Result. Excel shows an error alert.



Note: to enter an input message and error alert message, go to the Input Message and Error Alert tab.

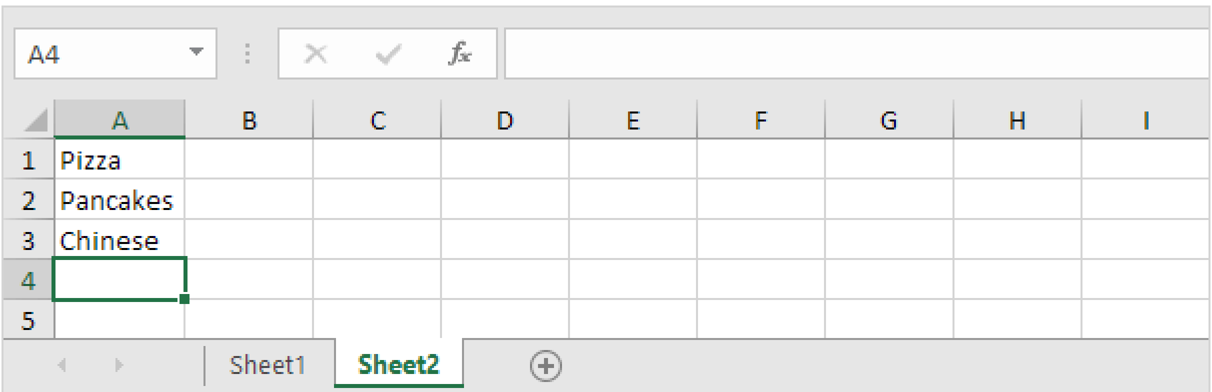
Drop-down List

Drop-down lists in Excel are helpful if you want to be sure that users select an item from a list, instead of typing their own values.

Create a Drop-down List

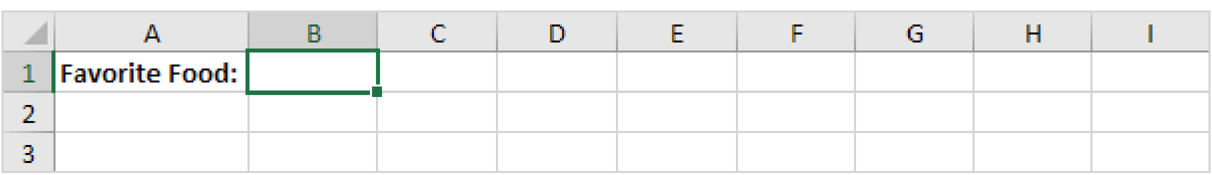
To create a drop-down list in Excel, execute the following steps.

1. On the second sheet, type the items you want to appear in the drop-down list.

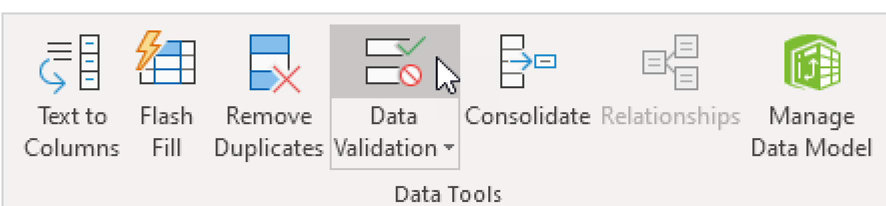


Note: if you don't want users to access the items on Sheet2, you can hide Sheet2. To achieve this, right click on the sheet tab of Sheet2 and click on Hide.

2. On the first sheet, select cell B1.



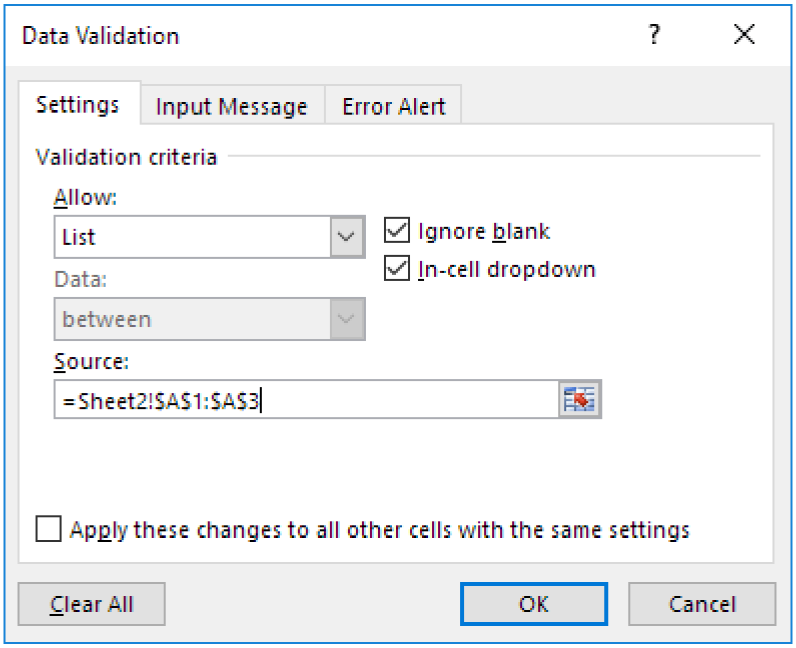
3. On the Data tab, in the Data Tools group, click Data Validation.



The 'Data Validation' dialog box appears.

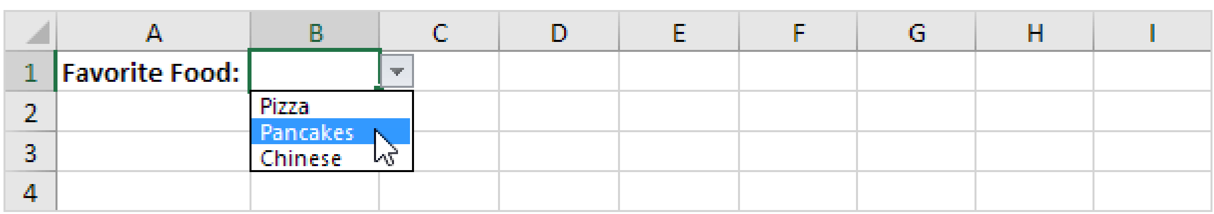
4. In the Allow box, click List.

5. Click in the Source box and select the range A1:A3 on Sheet2.



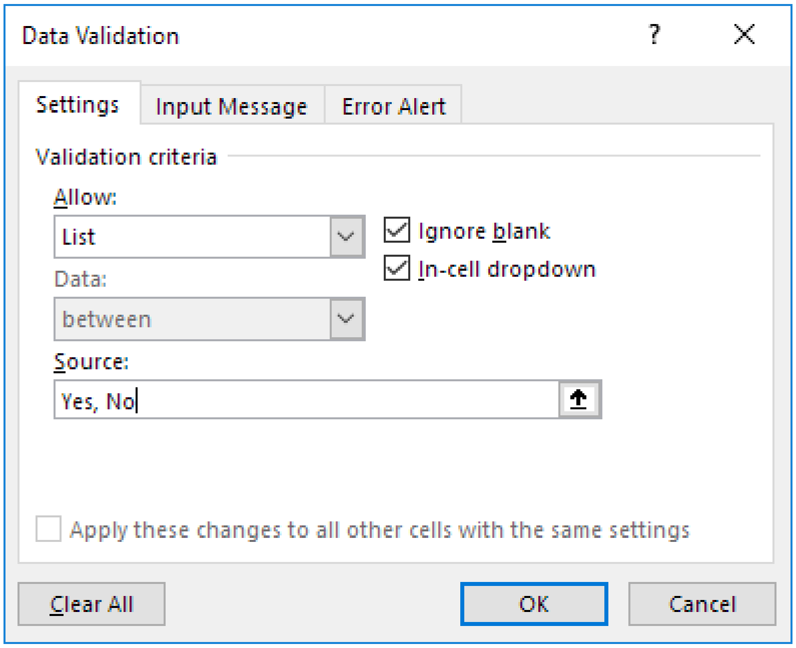
6. Click OK.

Result:



Note: to copy/paste a drop-down list, select the cell with the drop-down list and press CTRL + c, select another cell and press CTRL + v.

7. You can also type the items directly into the Source box, instead of using a range reference.

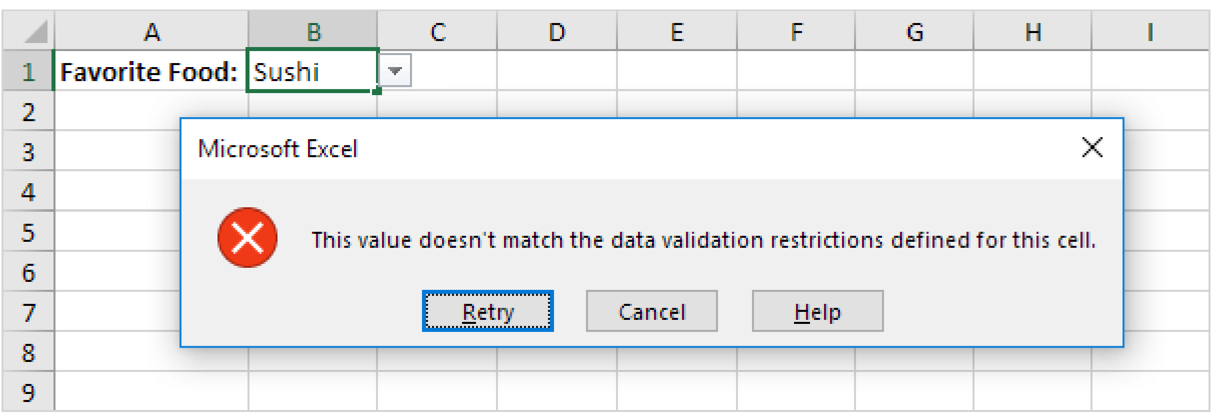


Note: this makes your drop-down list case sensitive. For example, if a user types yes, an error alert will be displayed.

Allow Other Entries

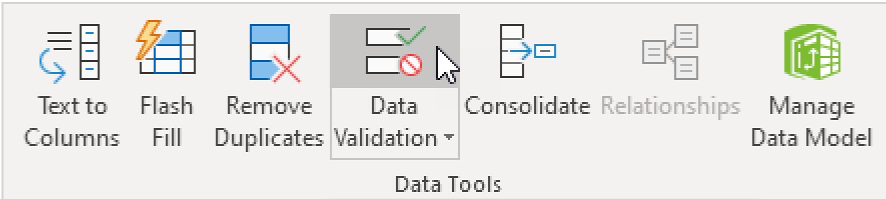
You can also create a drop-down list in Excel that allows other entries.

1. First, if you type a value that is not in the list, Excel shows an error alert.



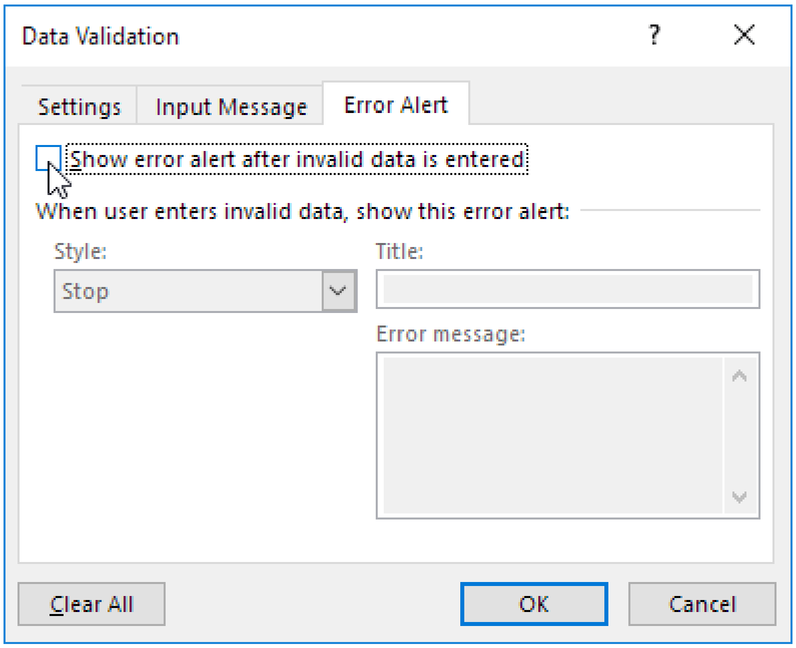
To allow other entries, execute the following steps.

2. On the Data tab, in the Data Tools group, click Data Validation.



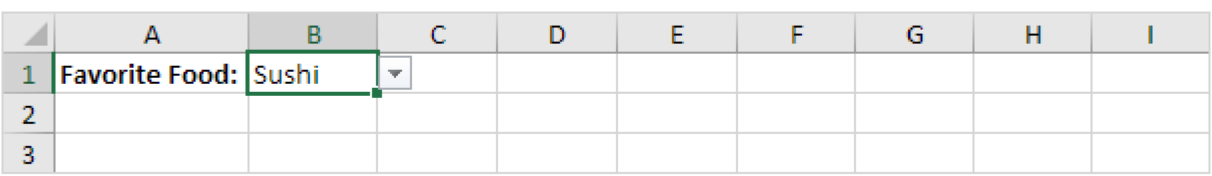
The 'Data Validation' dialog box appears.

3. On the Error Alert tab, uncheck 'Show error alert after invalid data is entered'.



4. Click OK.

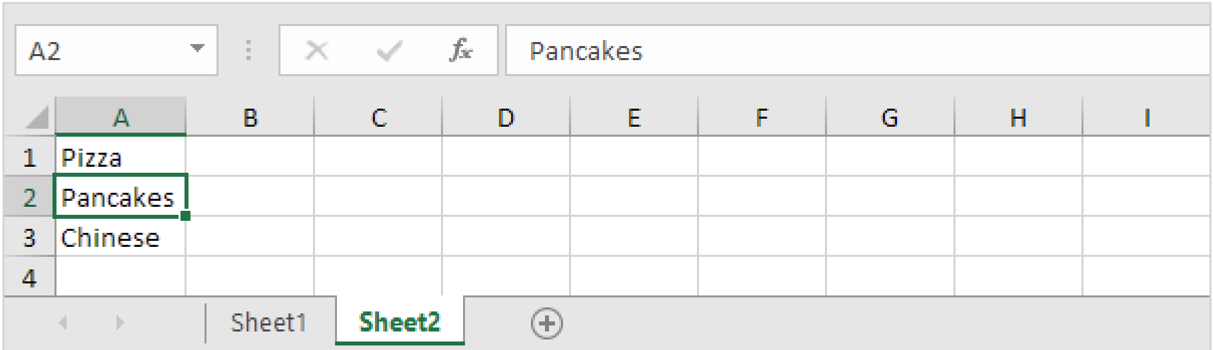
5. You can now enter a value that is not in the list.



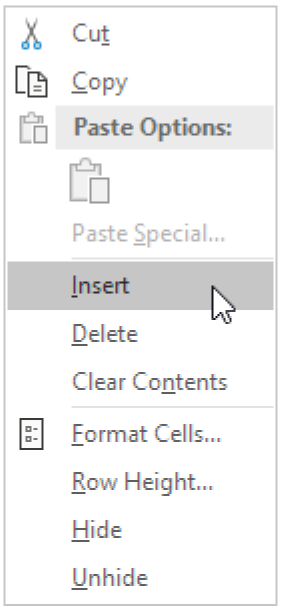
Add/Remove Items

You can add or remove items from a drop-down list in Excel without opening the 'Data Validation' dialog box and changing the range reference. This saves time.

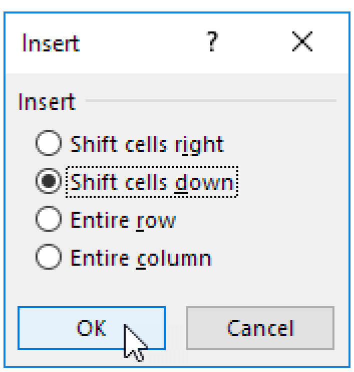
1. To add an item to a drop-down list, go to the items and select an item.



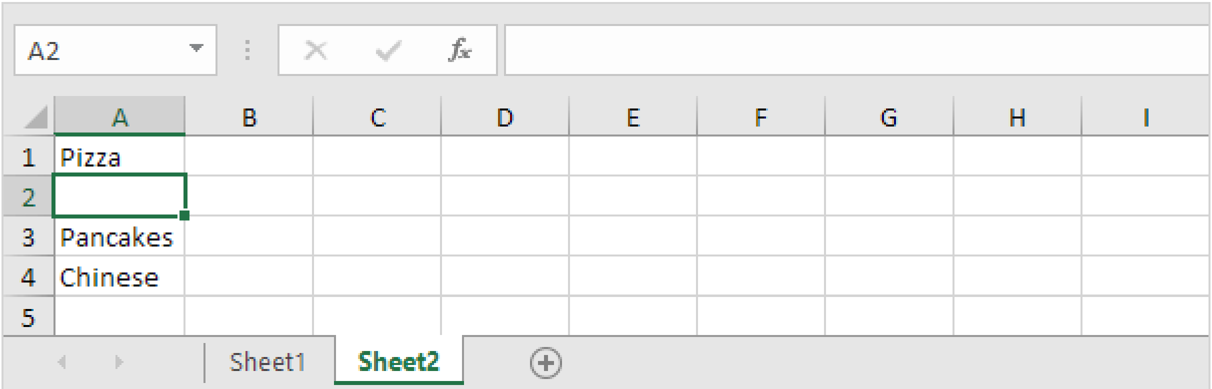
2. Right click, and then click Insert.



3. Select "Shift cells down" and click OK.

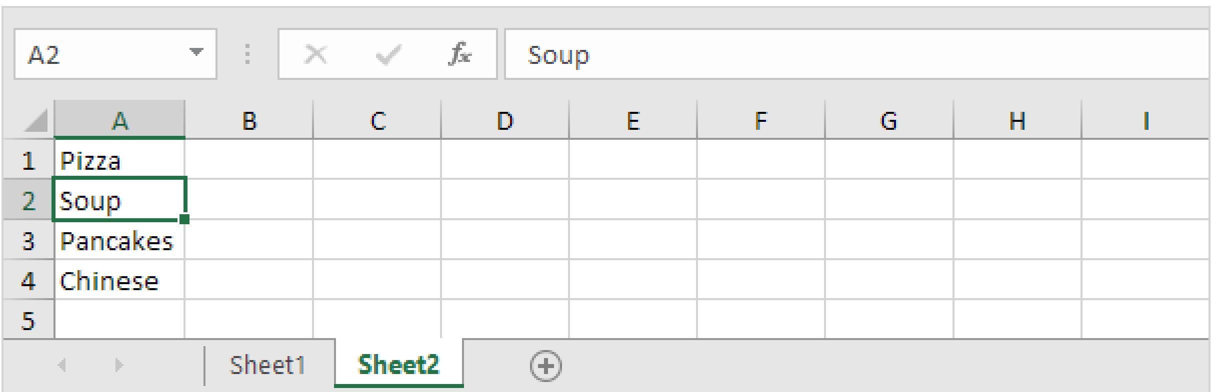


Result:

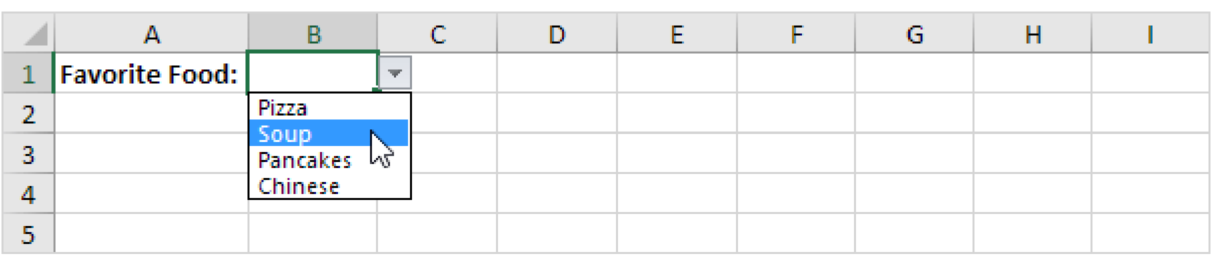


Note: Excel automatically changed the range reference from Sheet2!$A$1:$A$3 to Sheet2!$A$1:$A$4. You can check this by opening the 'Data Validation' dialog box.

4. Type a new item.



Result:

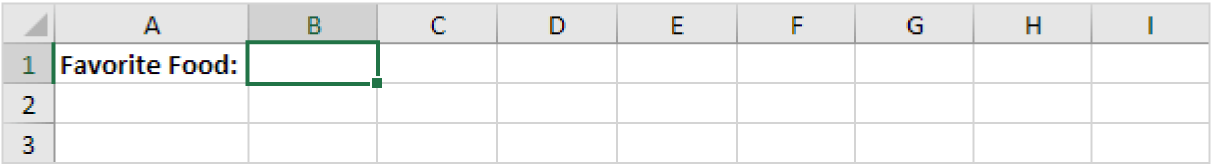


5. To remove an item from a drop-down list, at step 2, click Delete, select "Shift cells up" and click OK.

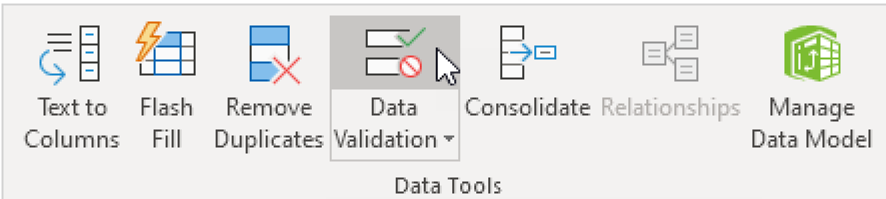
Dynamic Drop-down List

You can also use a formula that updates your drop-down list automatically when you add an item to the end of the list.

1. On the first sheet, select cell B1.



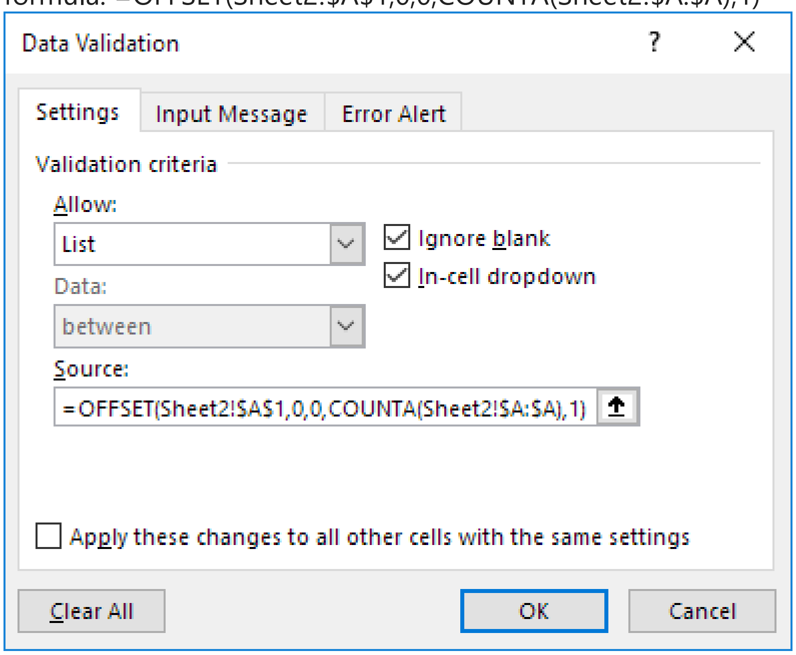
2. On the Data tab, in the Data Tools group, click Data Validation.



The 'Data Validation' dialog box appears.

3. In the Allow box, click List.

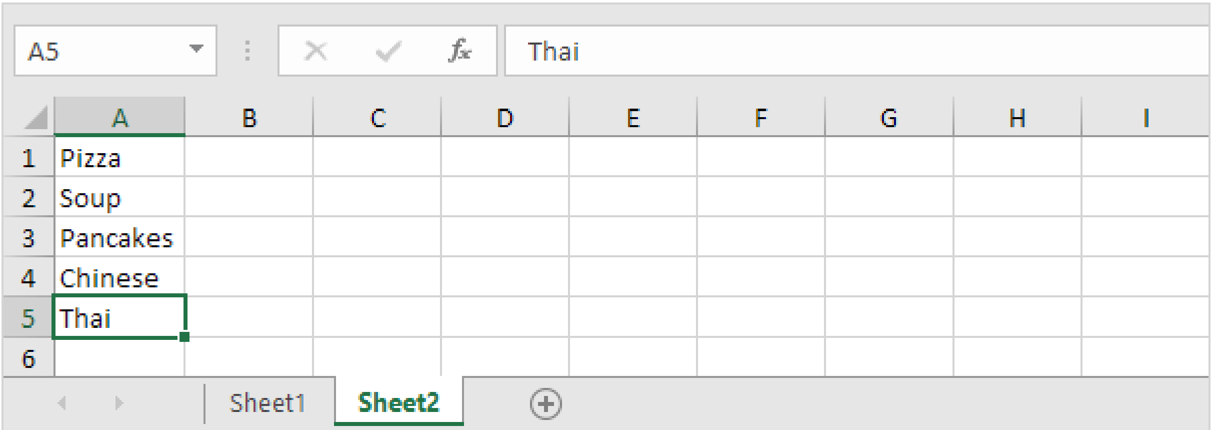
4. Click in the Source box and enter the formula: =OFFSET(Sheet2!$A$1,0,0,COUNTA(Sheet2!$A:$A),1)



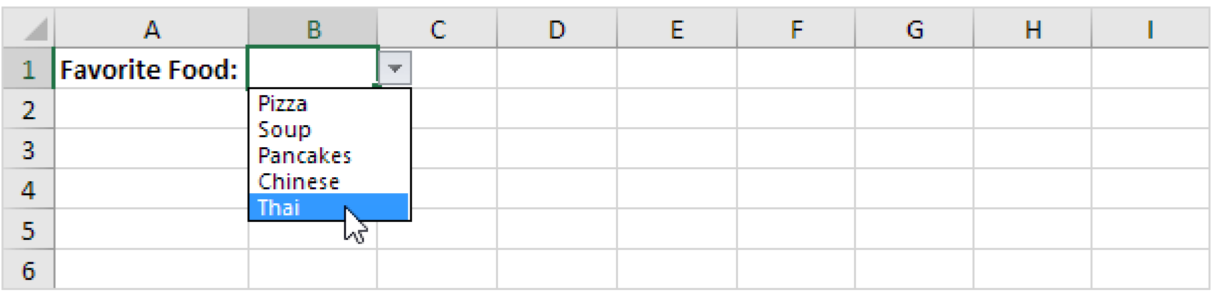
Explanation: the OFFSET function takes 5 arguments. Reference: Sheet2!$A$1, rows to offset: 0, columns to offset: 0, height: COUNTA(Sheet2!$A:$A) and width: 1. COUNTA(Sheet2!$A:$A) counts the number of values in column A on Sheet2 that are not empty. When you add an item to the list on Sheet2, COUNTA(Sheet2!$A:$A) increases. As a result, the range returned by the OFFSET function expands and the drop-down list will be updated.

5. Click OK.

6. On the second sheet, simply add a new item to the end of the list.



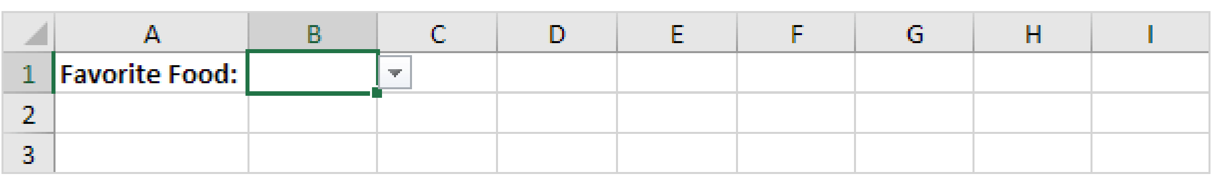
Result:



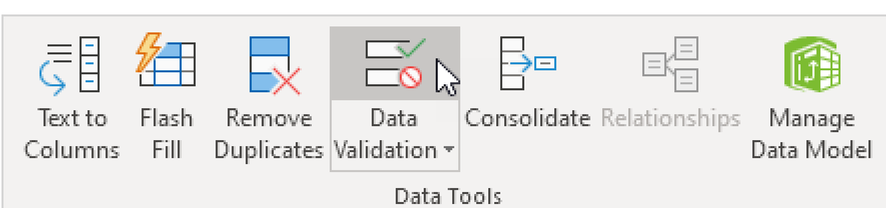
Remove a Drop-down List

To remove a drop-down list in Excel, execute the following steps.

1. Select the cell with the drop-down list.

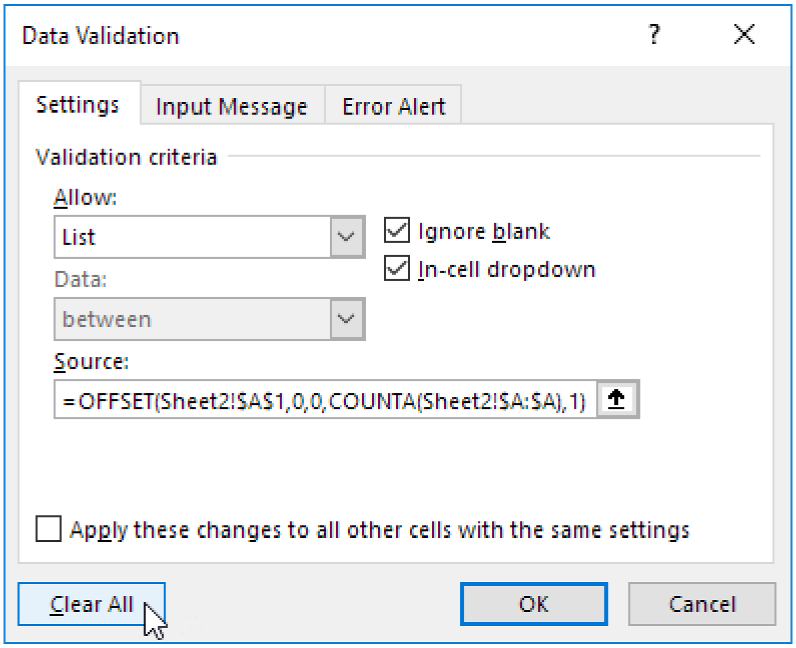


2. On the Data tab, in the Data Tools group, click Data Validation.



The 'Data Validation' dialog box appears.

3. Click Clear All.



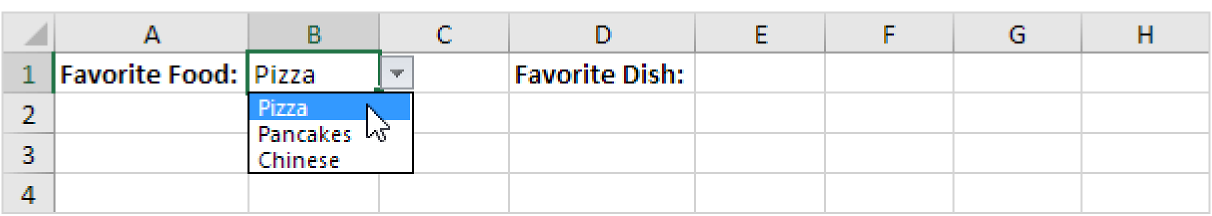
Note: to remove all other drop-down lists with the same settings, check "Apply these changes to all other cells with the same settings" before you click on Clear All.

4. Click OK.

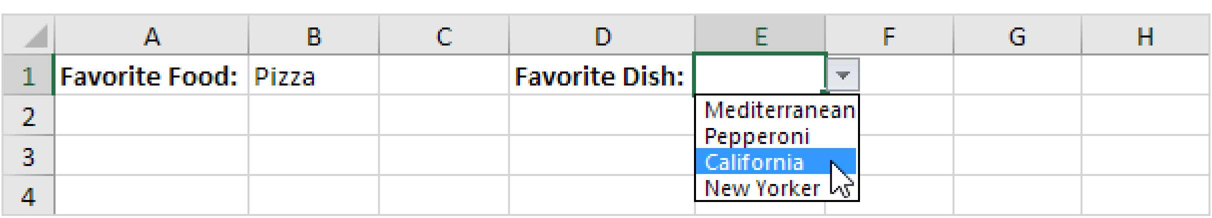
Dependent Drop-down Lists

Want to learn even more about drop-down lists in Excel? Learn how to create dependent drop-down lists.

1. For example, if the user selects Pizza from a first drop-down list.



2. A second drop-down list contains the Pizza items.



3. But if the user selects Chinese from the first drop-down list, the second drop-down list contains the Chinese dishes.

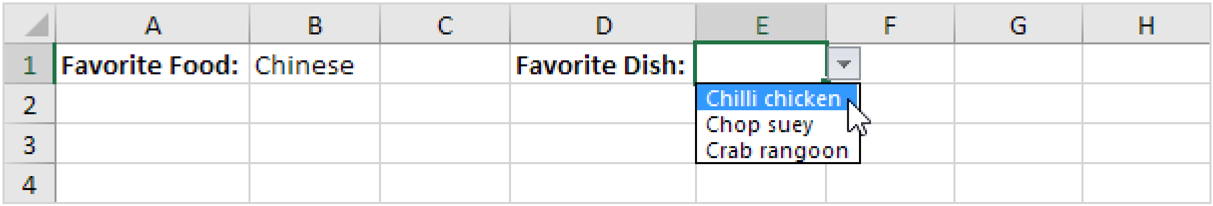
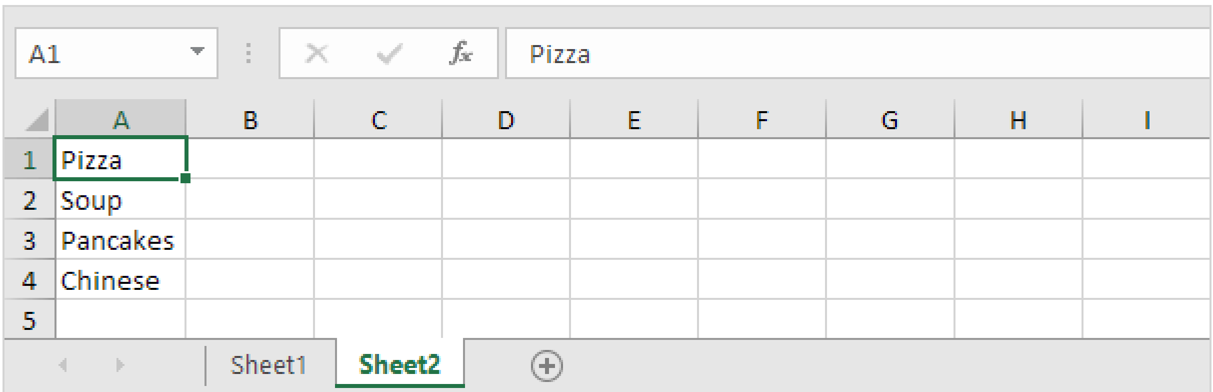


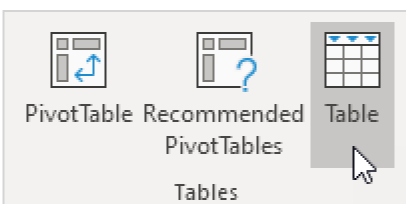
Table Magic

You can also store your items in an Excel table to create a dynamic drop-down list.

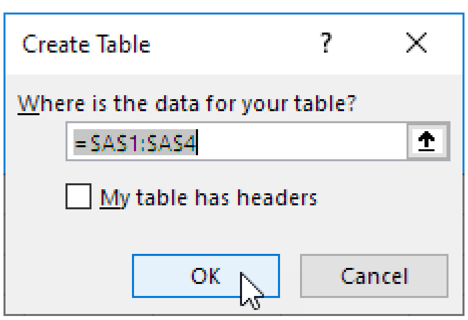
1. On the second sheet, select a list item.



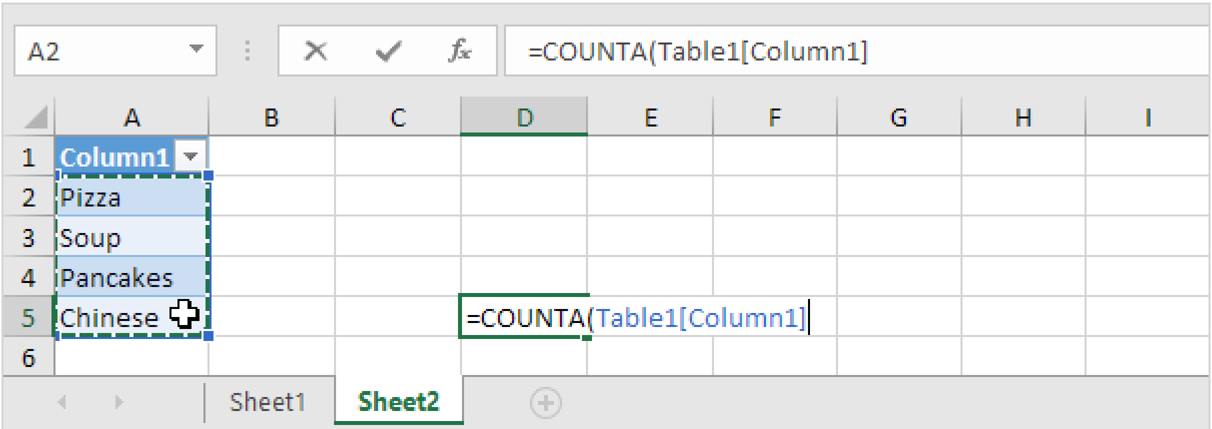
2. On the Insert tab, in the Tables group, click Table.



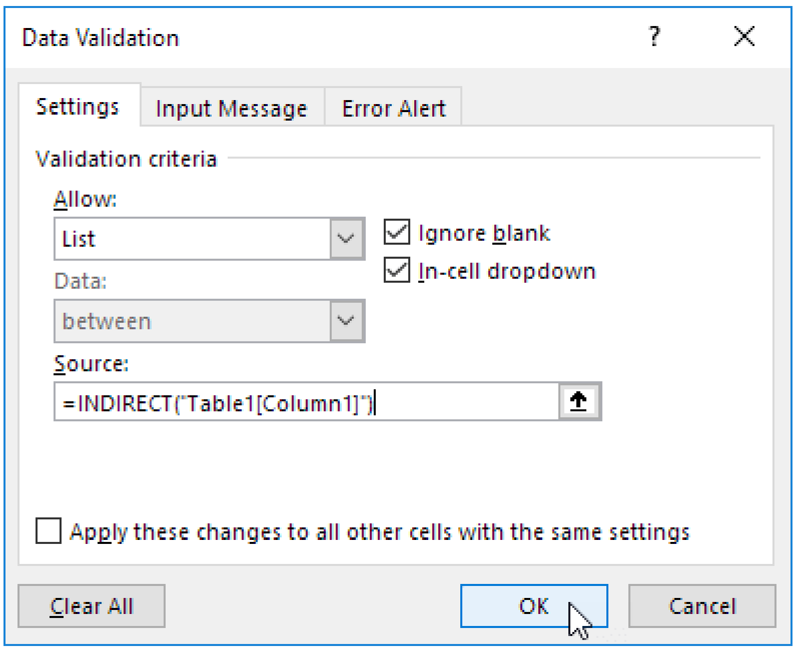
3. Excel automatically selects the data for you. Click OK.



4. If you select the list, Excel reveals the structured reference.

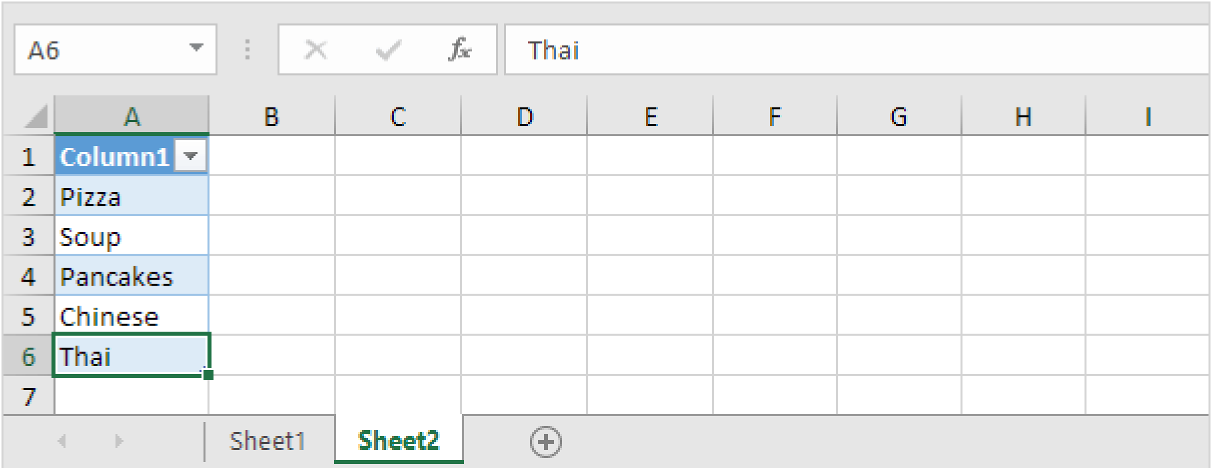


5. Use this structured reference to create a dynamic drop-down list.

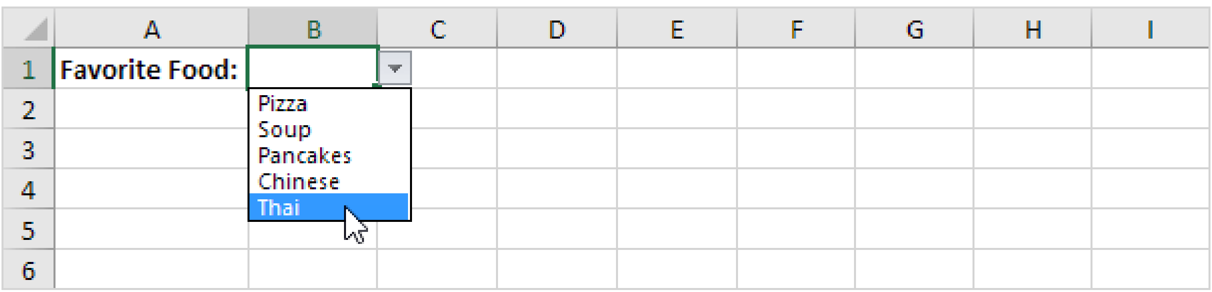


Explanation: the INDIRECT function in Excel converts a text string into a valid reference.

6. On the second sheet, simply add a new item to the end of the list.

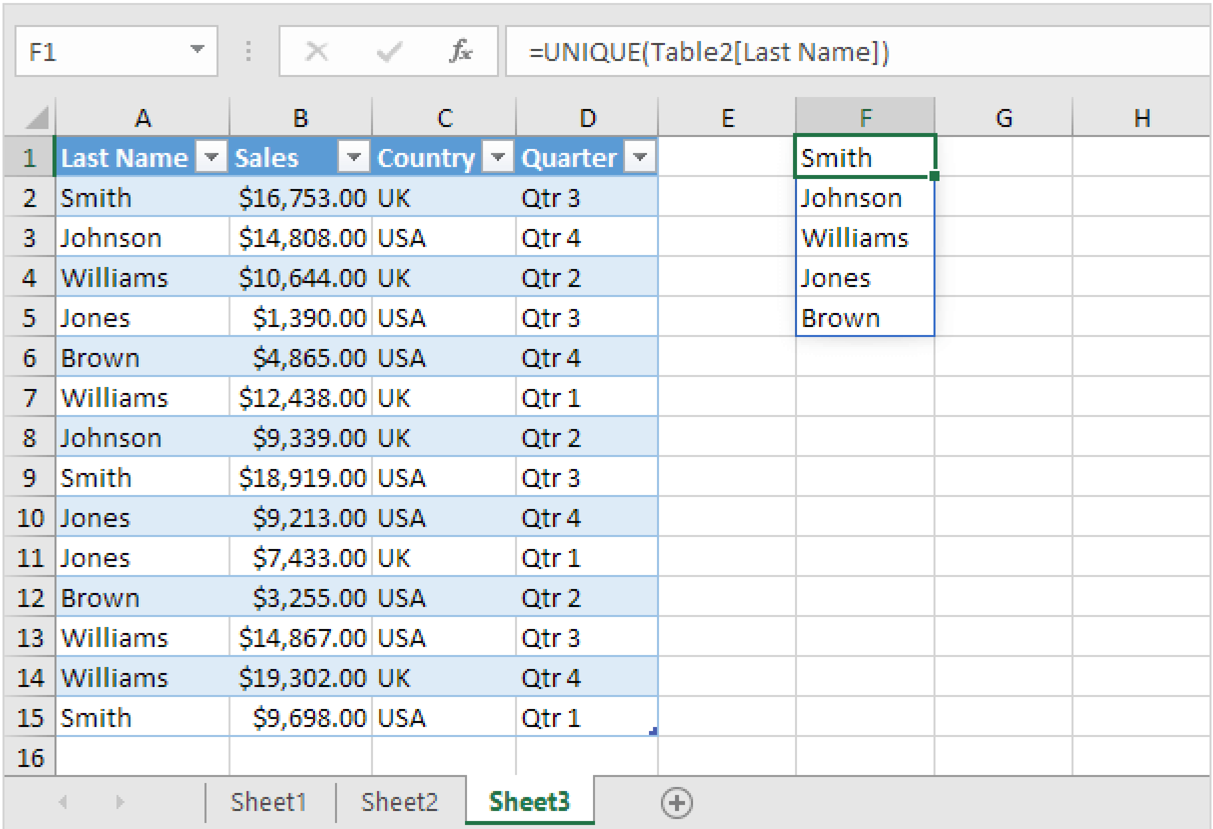


Result:



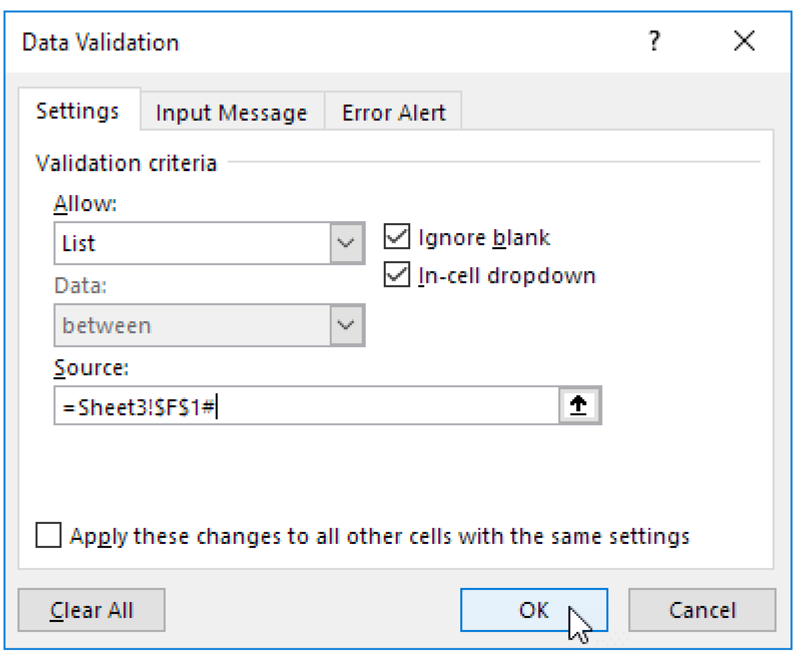
Note: try it yourself. Download the Excel file and create this drop-down list.

7. When using tables, use the UNIQUE function in Excel 365 to extract unique list items.



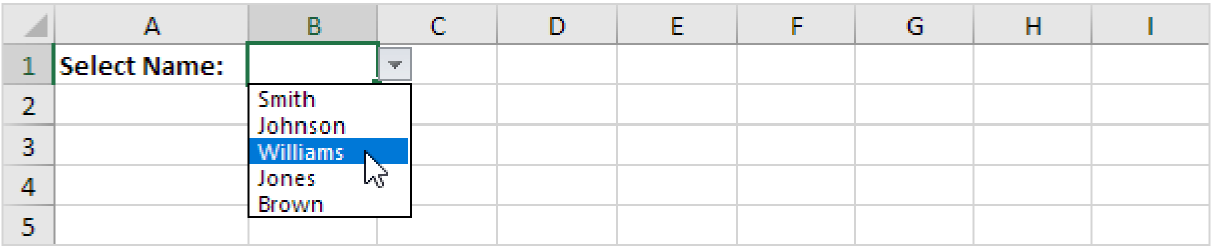
Note: this dynamic array function, entered into cell F1, fills multiple cells. Wow! This behavior in Excel 365 is called spilling.

8. Use this spill range to create a magic drop-down list.



Explanation: always use the first cell (F1) and a hash character to refer to a spill range.

Result:

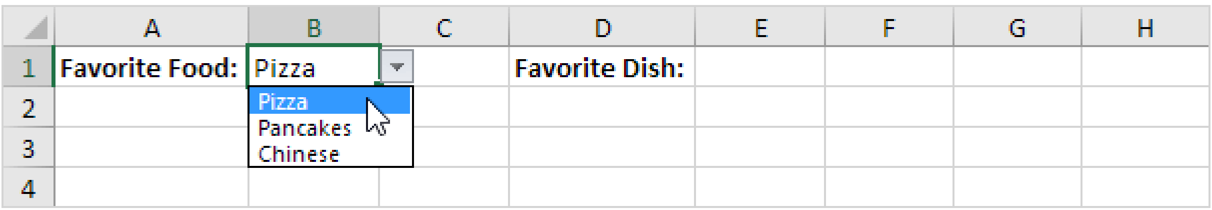


Note: when you add new records, the UNIQUE function automatically extracts new unique list items and Excel automatically updates the drop-down list.

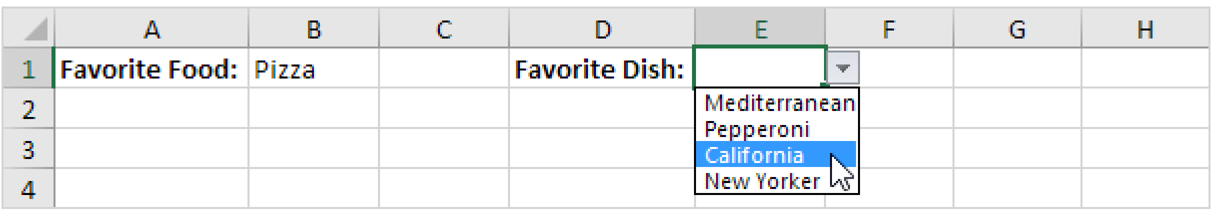
Dependent Drop-down Lists

This example describes how to create dependent drop-down lists in Excel. Here's what we are trying to achieve:

The user selects Pizza from a drop-down list.



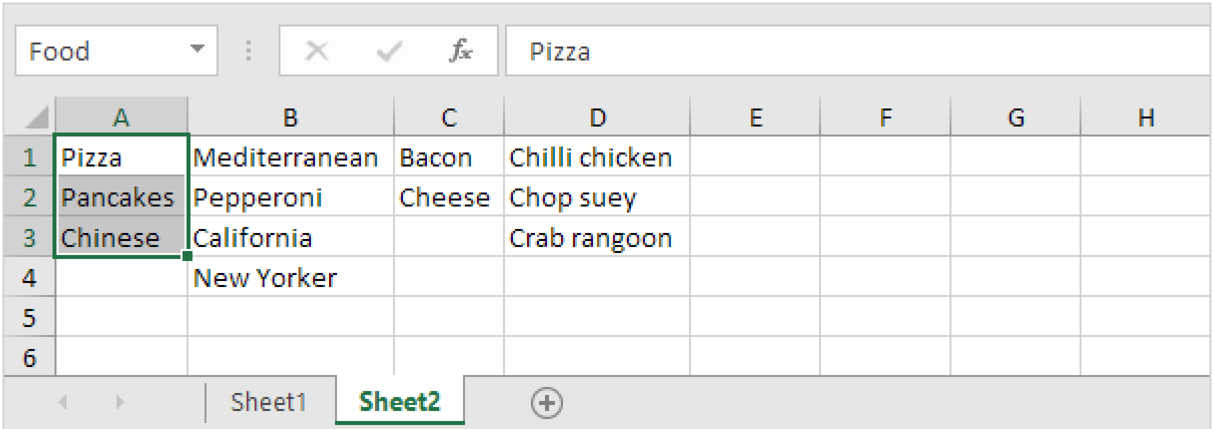
As a result, a second drop-down list contains the Pizza items.



To create these dependent drop-down lists, execute the following steps.

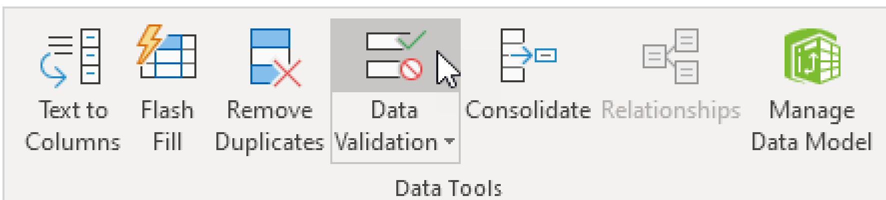
1. On the second sheet, create the following named ranges.

Name	Range Address
Food	A1:A3
Pizza	B1:B4
Pancakes	C1:C2
Chinese	D1:D3



2. On the first sheet, select cell B1.

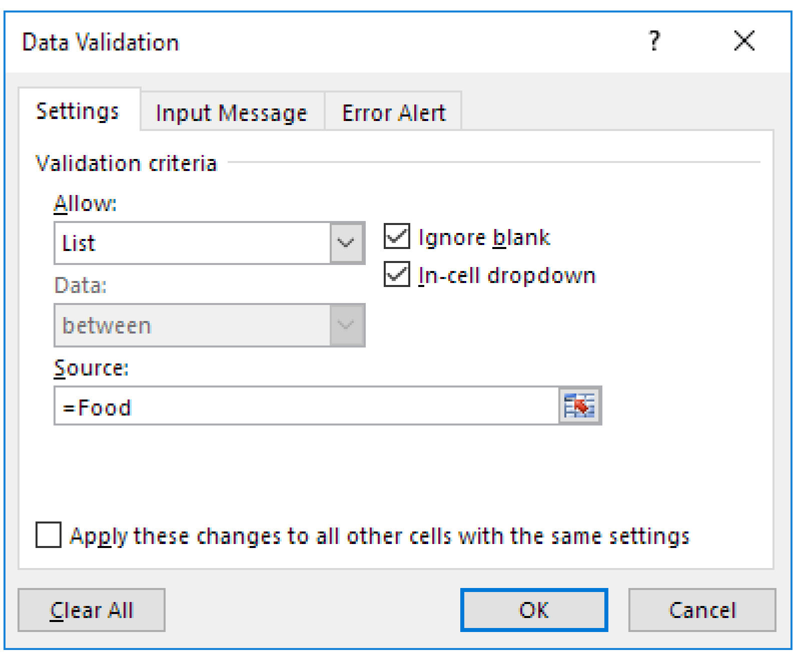
3. On the Data tab, in the Data Tools group, click Data Validation.



The 'Data Validation' dialog box appears.

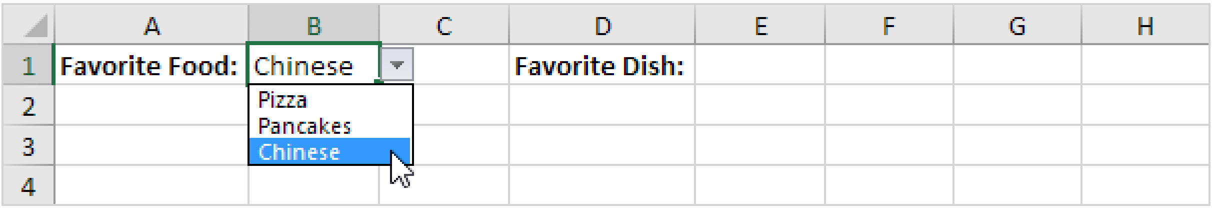
4. In the Allow box, click List.

5. Click in the Source box and type =Food.



6. Click OK.

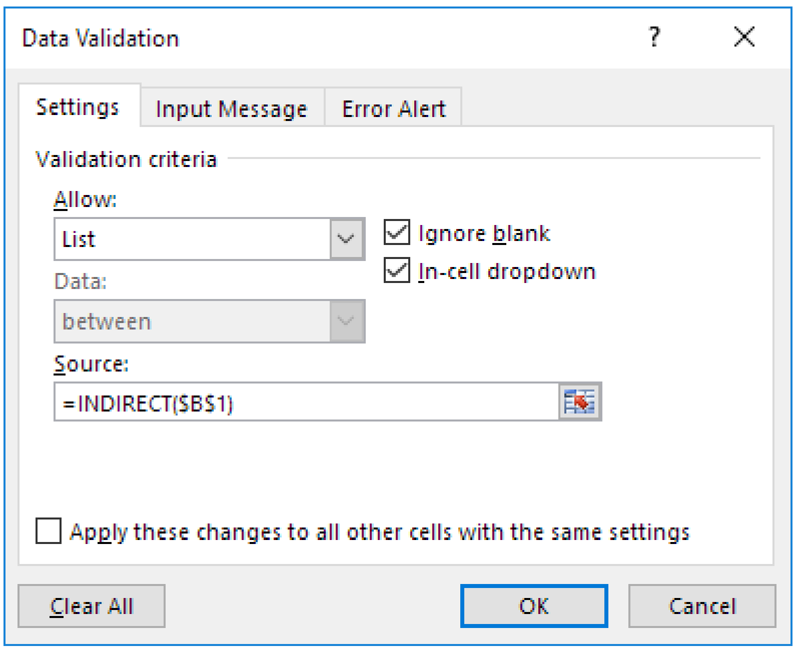
Result:



7. Next, select cell E1.

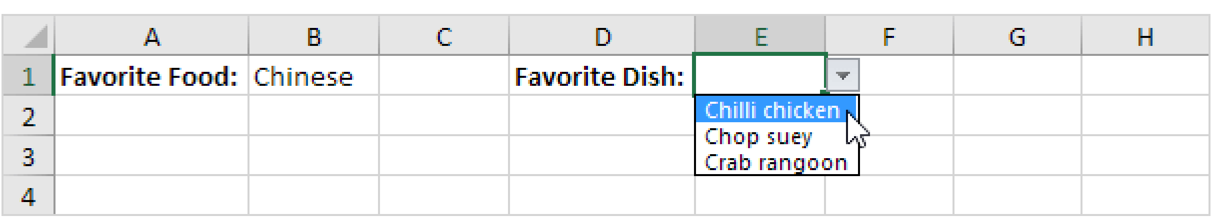
8. In the Allow box, click List.

9. Click in the Source box and type =INDIRECT($B$1).



10. Click OK.

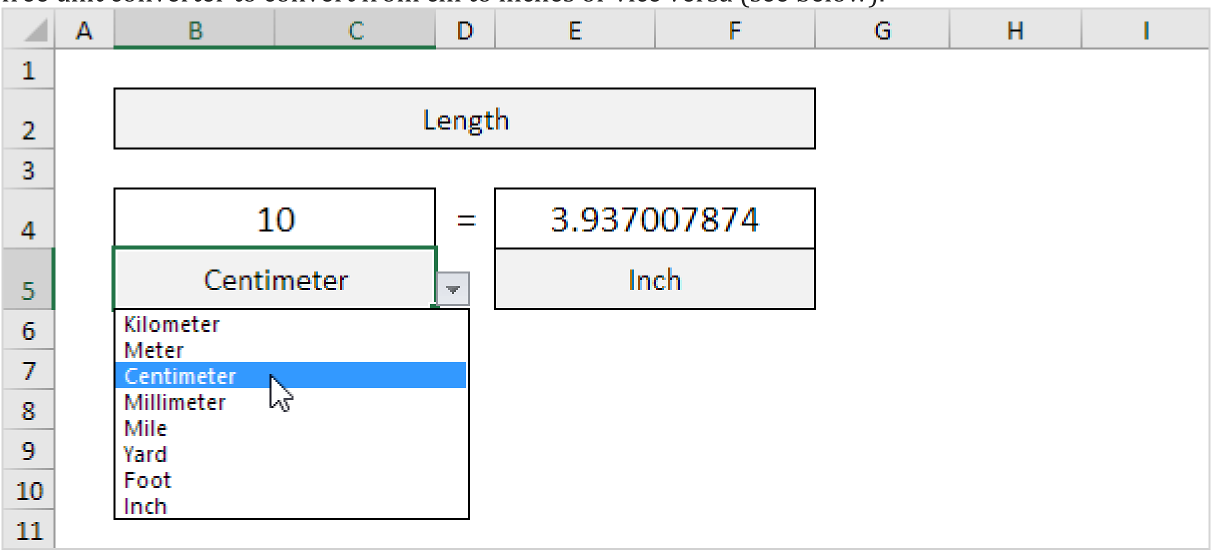
Result:



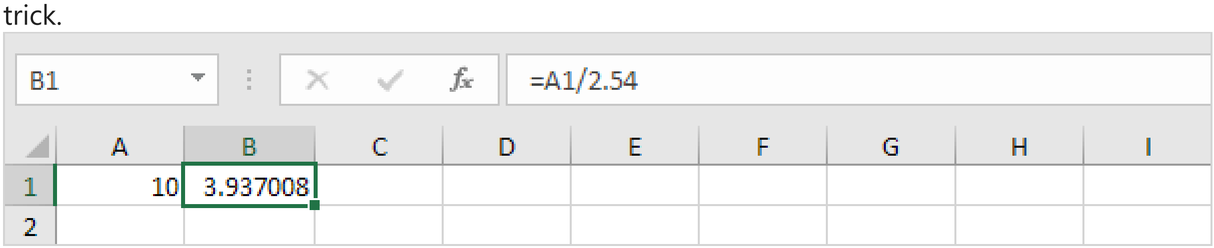
Explanation: the INDIRECT function returns the reference specified by a text string. For example, the user selects Chinese from the first drop-down list. =INDIRECT($B$1) returns the Chinese reference. As a result, the second drop-down lists contains the Chinese items.

Cm to inches

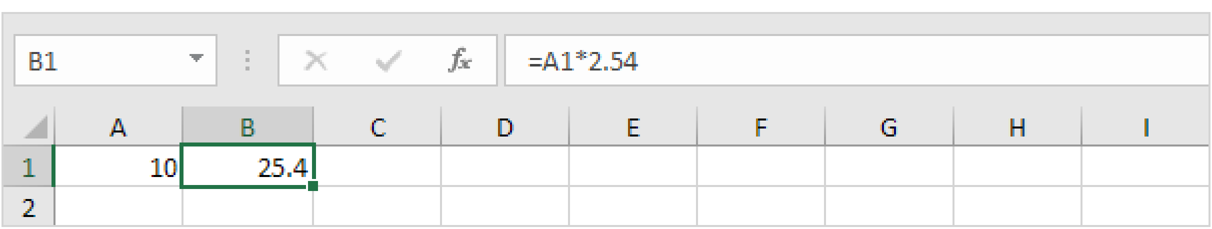
1 cm = 0.3937 inch and 1 inch = 2.54 cm. Use a simple formula, the CONVERT function or download our free unit converter to convert from cm to inches or vice versa (see below).



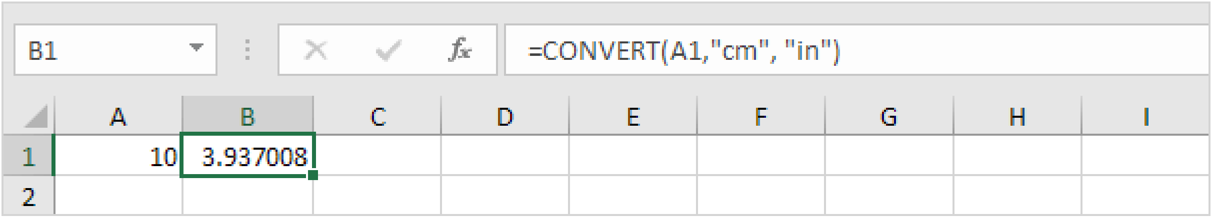
1. First, to convert from centimeters to inches, divide by 2.54. The simple formula below does the trick.



2. To convert from inches to centimeters, multiply by 2.54.

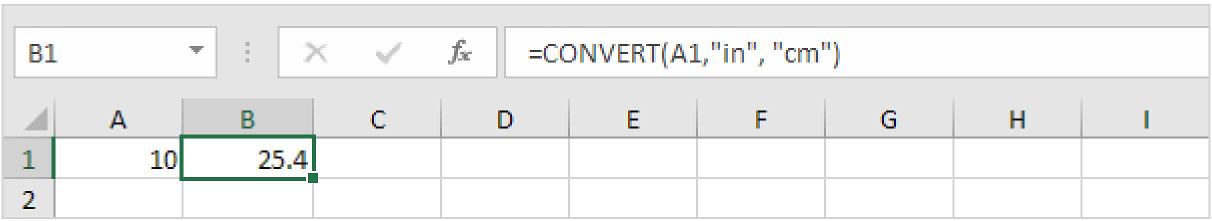


3. You can also use the CONVERT function in Excel to convert from centimeters to inches.



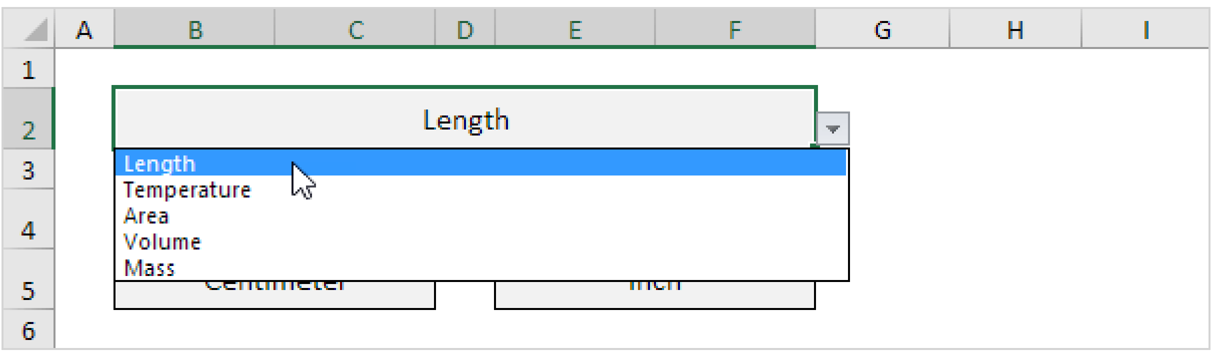
Note: the CONVERT function has three arguments (number, from_unit and to_unit).

4. You can also use the CONVERT function in Excel to convert from inches to centimeters.



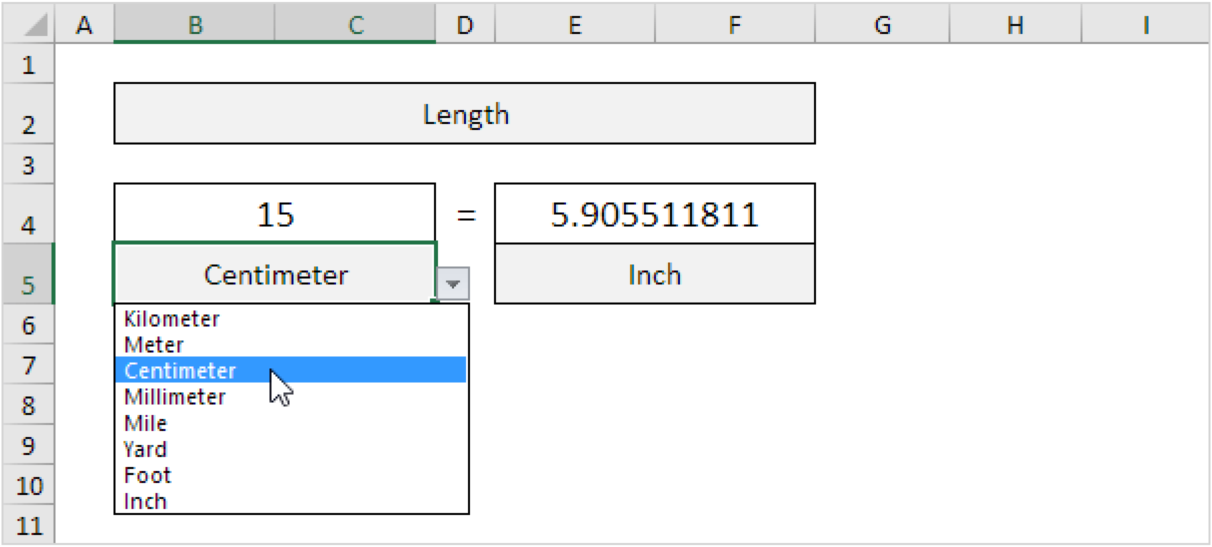
Download our free unit converter to quickly and easily convert from cm to inches (or vice versa).

5. Select Length from the category drop-down list.



Explanation: because you selected Length from this drop-down list, the dependent drop-down lists in cell B5 and cell E5 now contain length units.

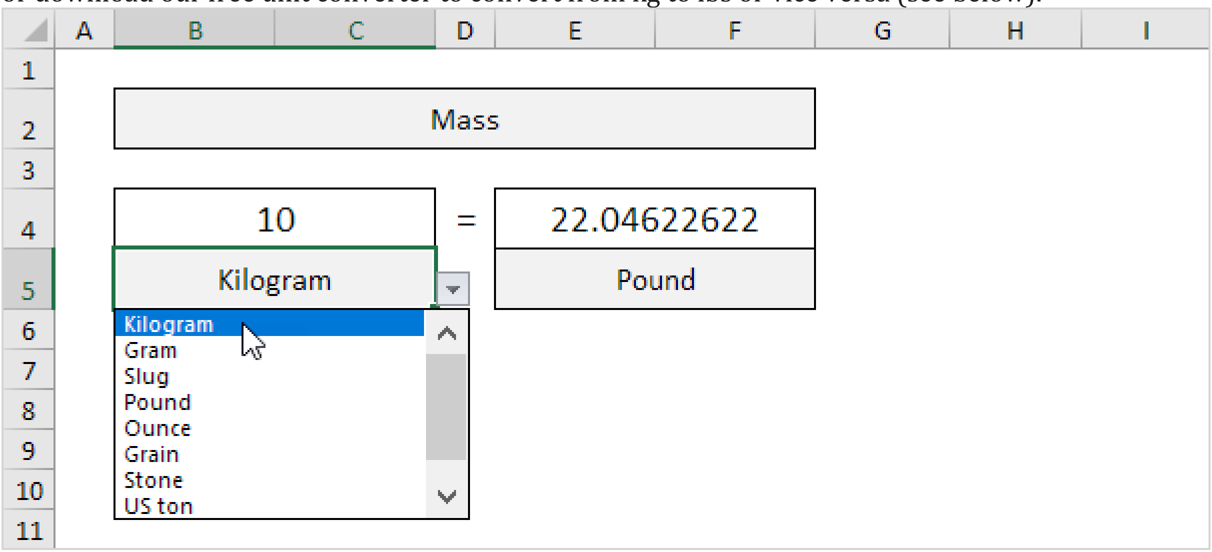
6. Enter a number and select the correct length units.



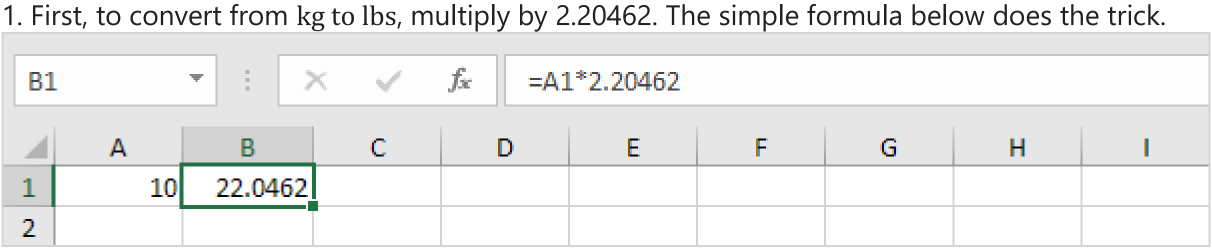
Explanation: the unit converter uses worksheet change events to automatically execute VBA code when you change a cell (B2, B4, B5, E4 or E5). The VBA code uses the CONVERT function to convert from one measurement unit to another and uses the VLOOKUP function to lookup the correct unit abbreviations (cm, in, yd, ft, etc.) stored on the second worksheet.

Kg to lbs

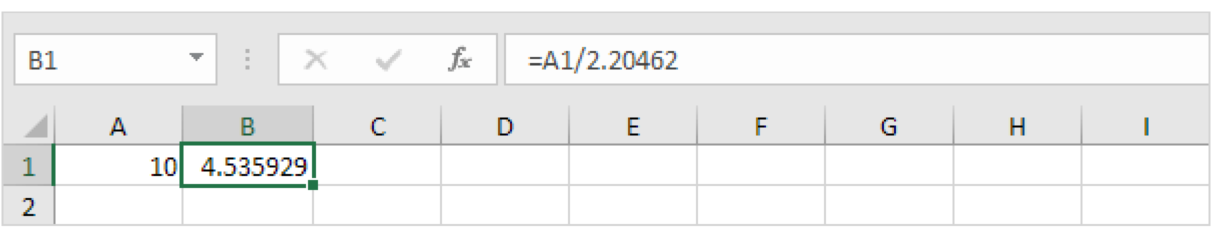
1 kg = 2.20462 lbs (pounds) and 1 lb (pound) = 0.45359 kg. Use a simple formula, the CONVERT function or download our free unit converter to convert from kg to lbs or vice versa (see below).



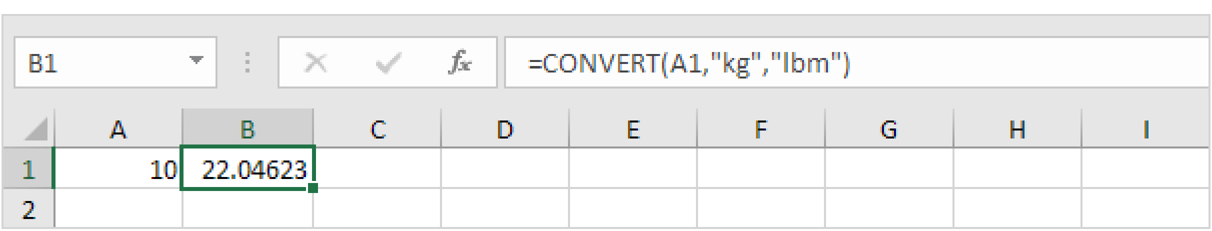
1. First, to convert from kg to lbs, multiply by 2.20462. The simple formula below does the trick.



2. To convert from lbs to kg, divide by 2.20462.

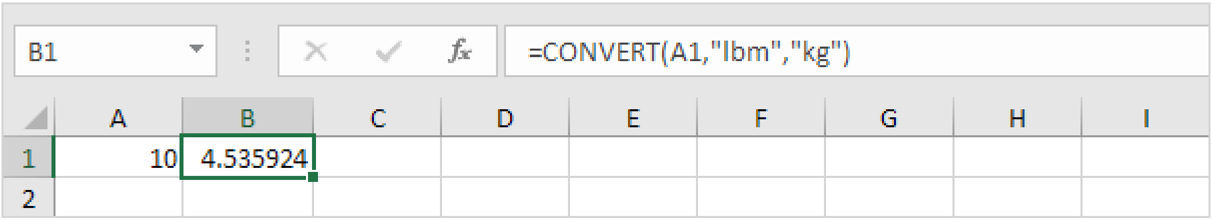


3. You can also use the CONVERT function in Excel to convert from kg to lbs.



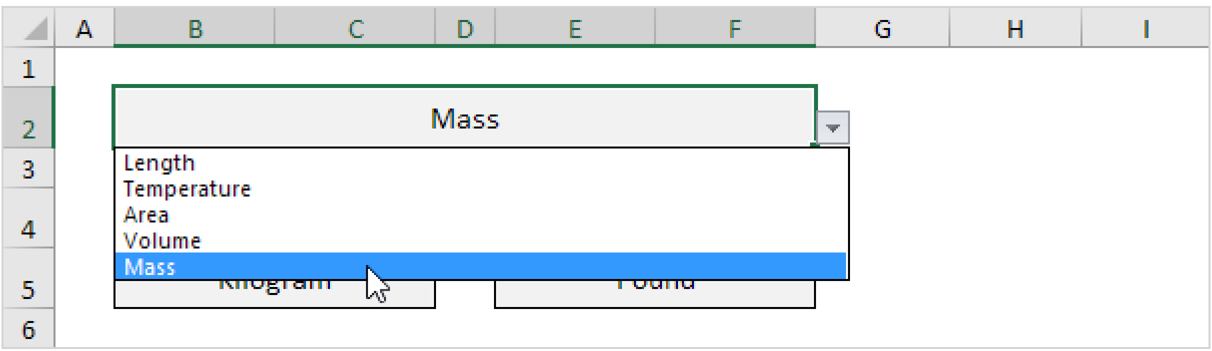
Note: the CONVERT function has three arguments (number, from_unit and to_unit). The unit abbreviation lbm stands for pound mass.

4. You can also use the CONVERT function in Excel to convert from lbs to kg.



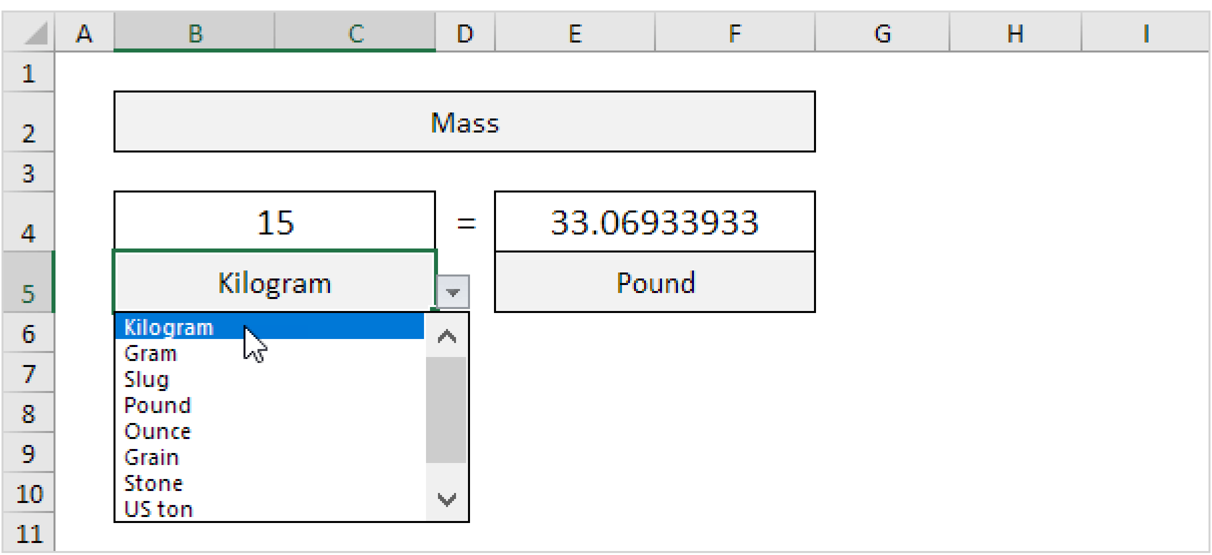
Download our free unit converter to quickly and easily convert from kg to lbs (or vice versa).

5. Select Mass from the category drop-down list.



Explanation: because you selected Mass from this drop-down list, the dependent drop-down lists in cell B5 and cell E5 now contain mass units.

6. Enter a number and select the correct mass units.



Explanation: the unit converter uses worksheet change events to automatically execute VBA code when you change a cell (B2, B4, B5, E4 or E5). The VBA code uses the CONVERT function to convert from one measurement unit to another and uses the VLOOKUP function to lookup the correct unit abbreviations (kg, g, lbm, ozm, etc.) stored on the second worksheet.